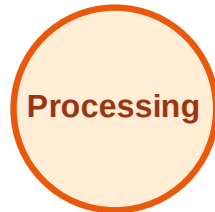


BFS Traversal

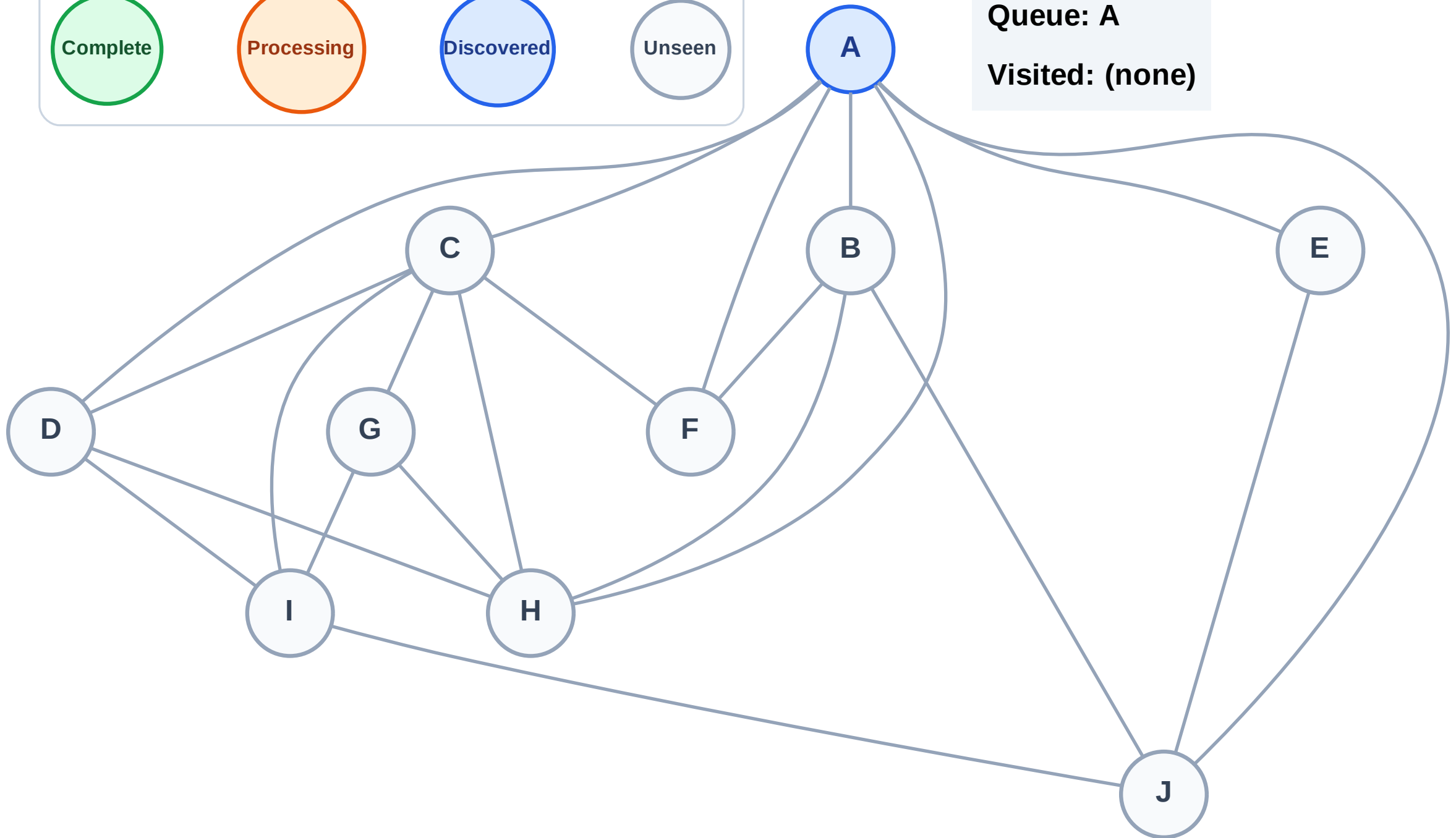
Step 1: Start at A

Legend



Queue: A

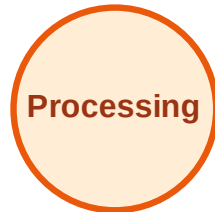
Visited: (none)



BFS Traversal

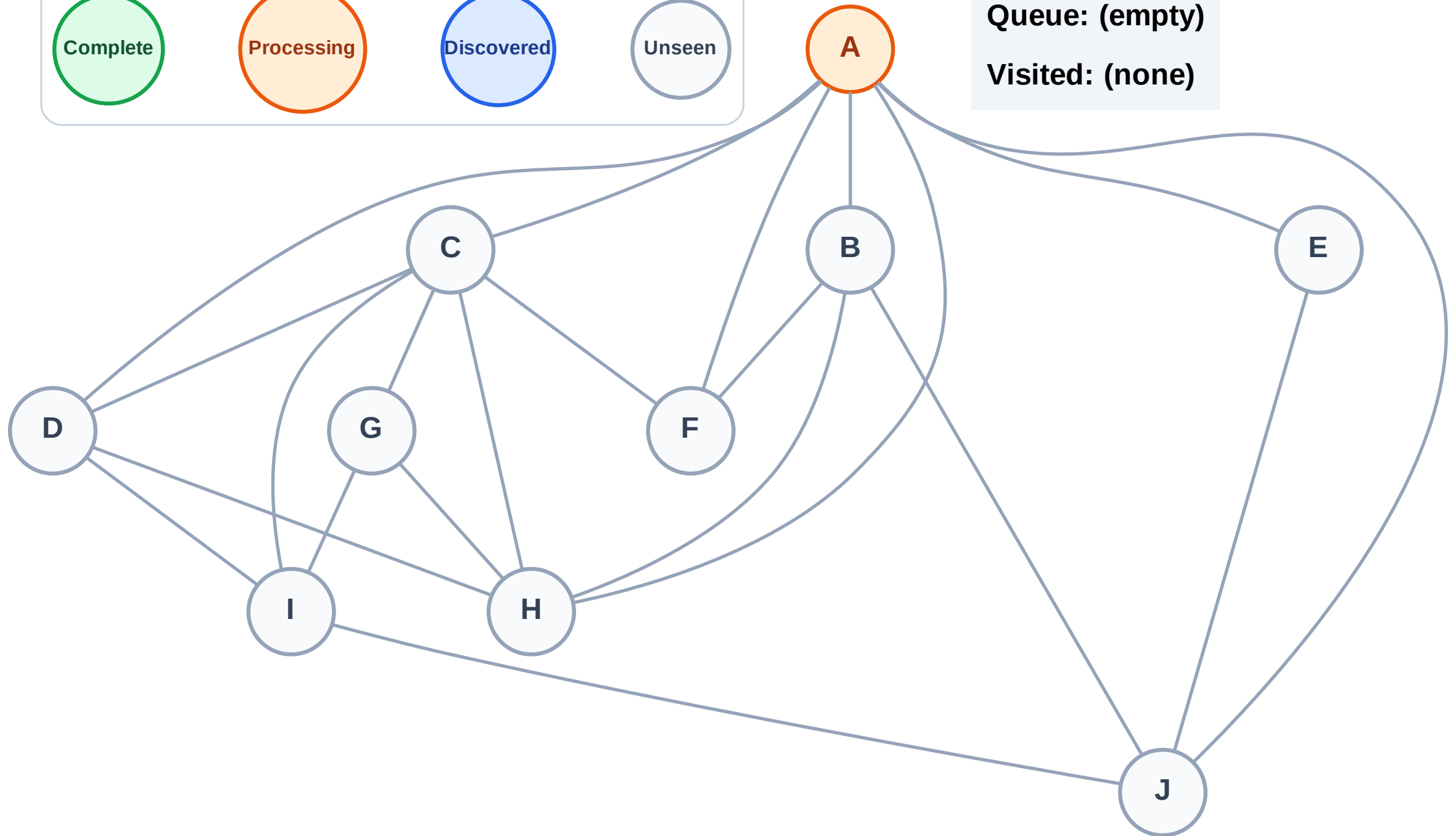
Step 2: Process node A

Legend



Queue: (empty)

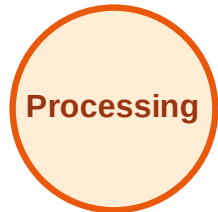
Visited: (none)



BFS Traversal

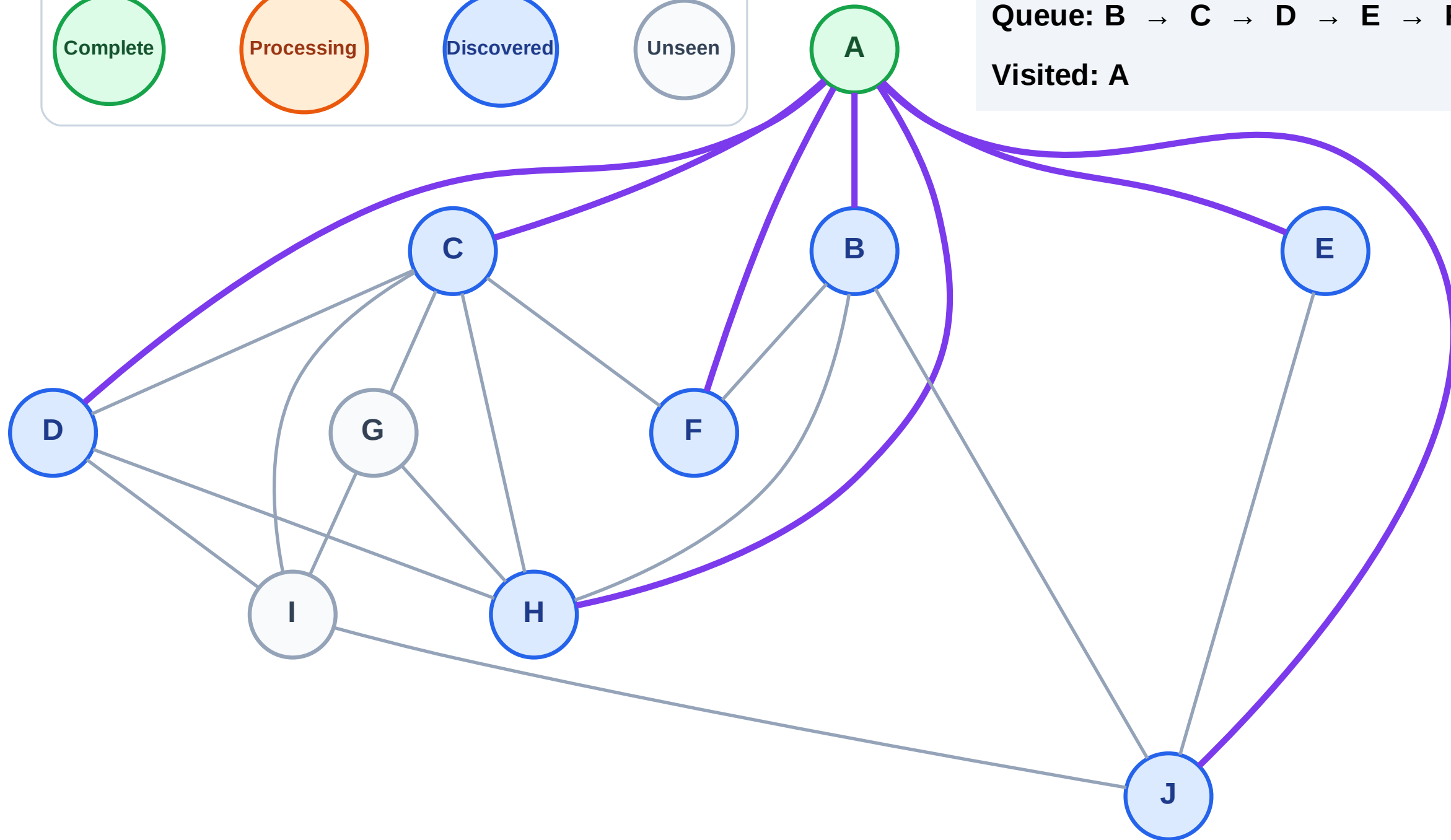
Step 3: Discover B, C, D, E, F, H, J from A

Legend



Queue: B → C → D → E → F → H → J

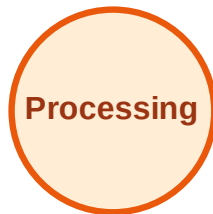
Visited: A



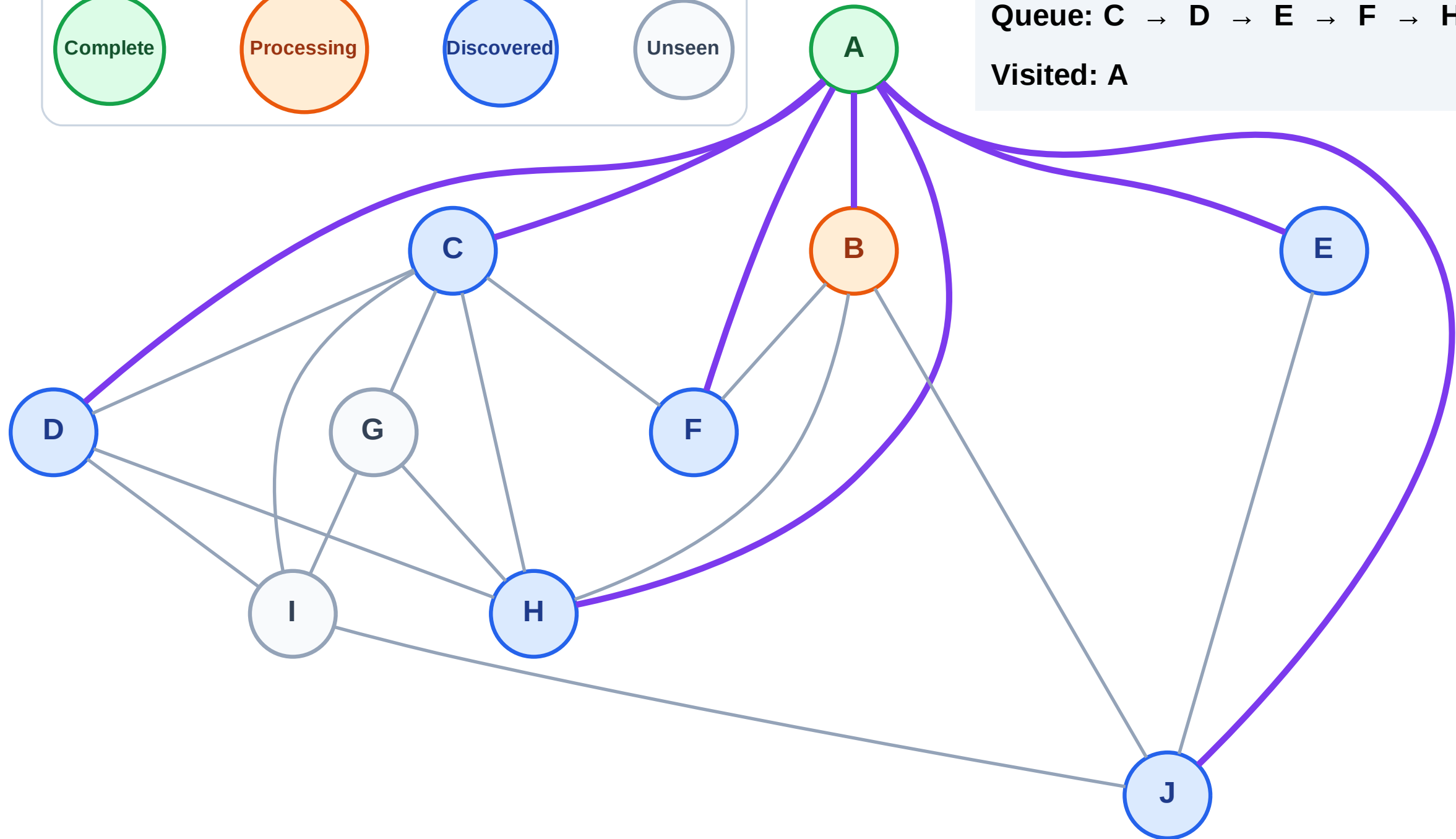
BFS Traversal

Step 4: Process node B

Legend



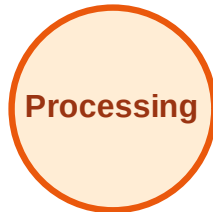
Queue: C → D → E → F → H → J
Visited: A



BFS Traversal

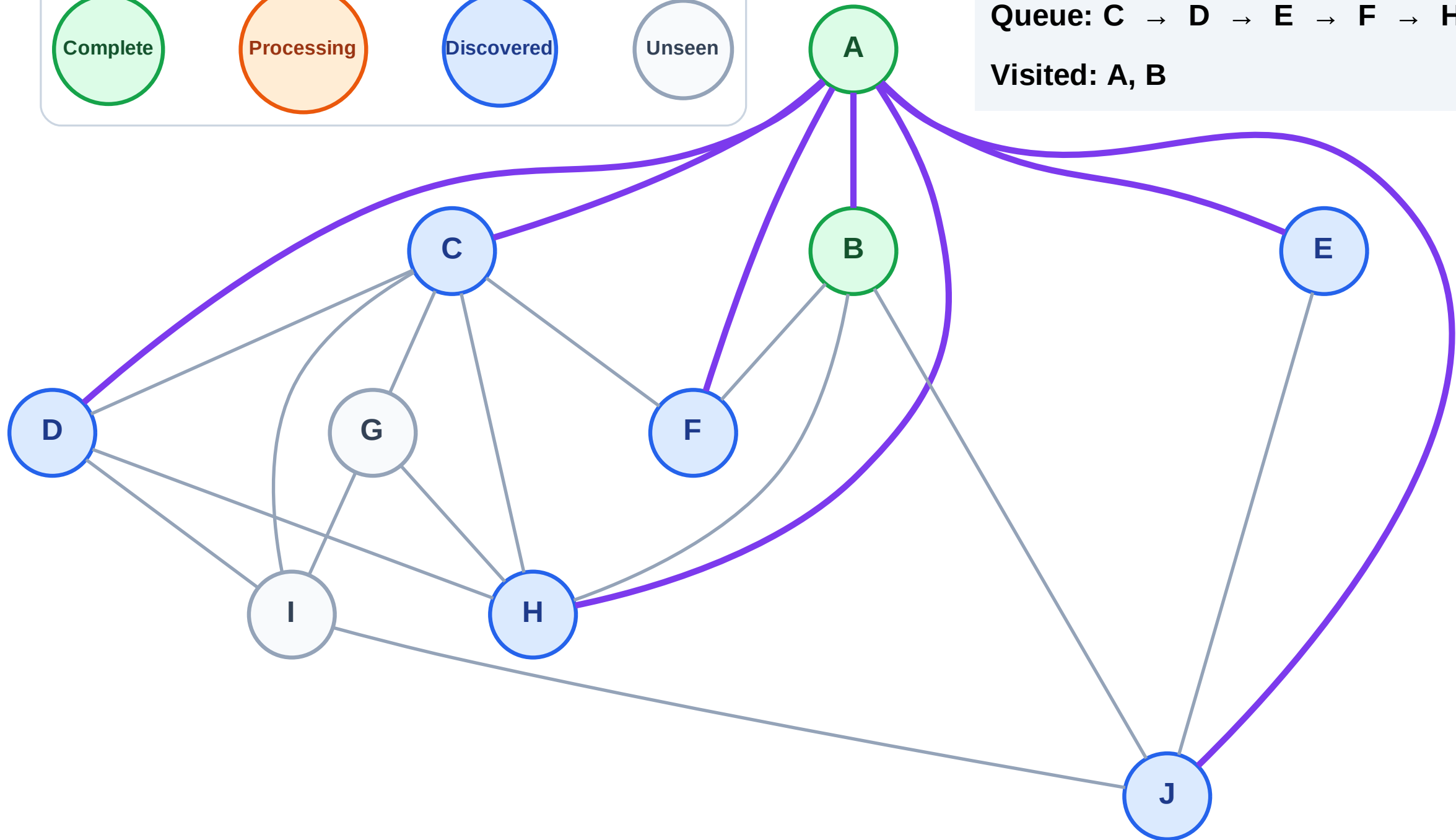
Step 5: No new neighbors from B

Legend



Queue: C → D → E → F → H → J

Visited: A, B



BFS Traversal

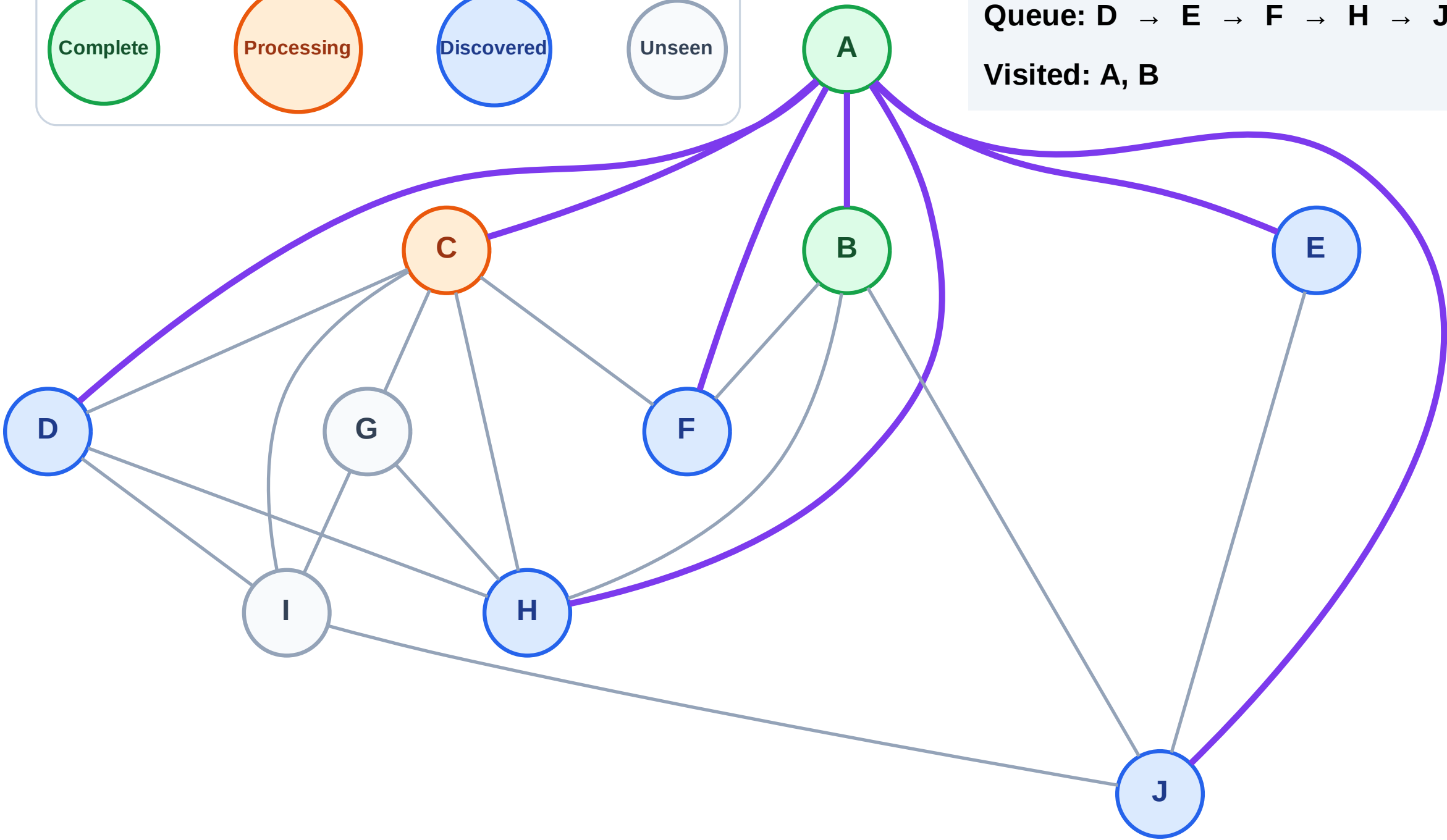
Step 6: Process node C

Legend



Queue: D → E → F → H → J

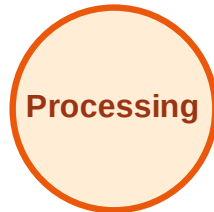
Visited: A, B



BFS Traversal

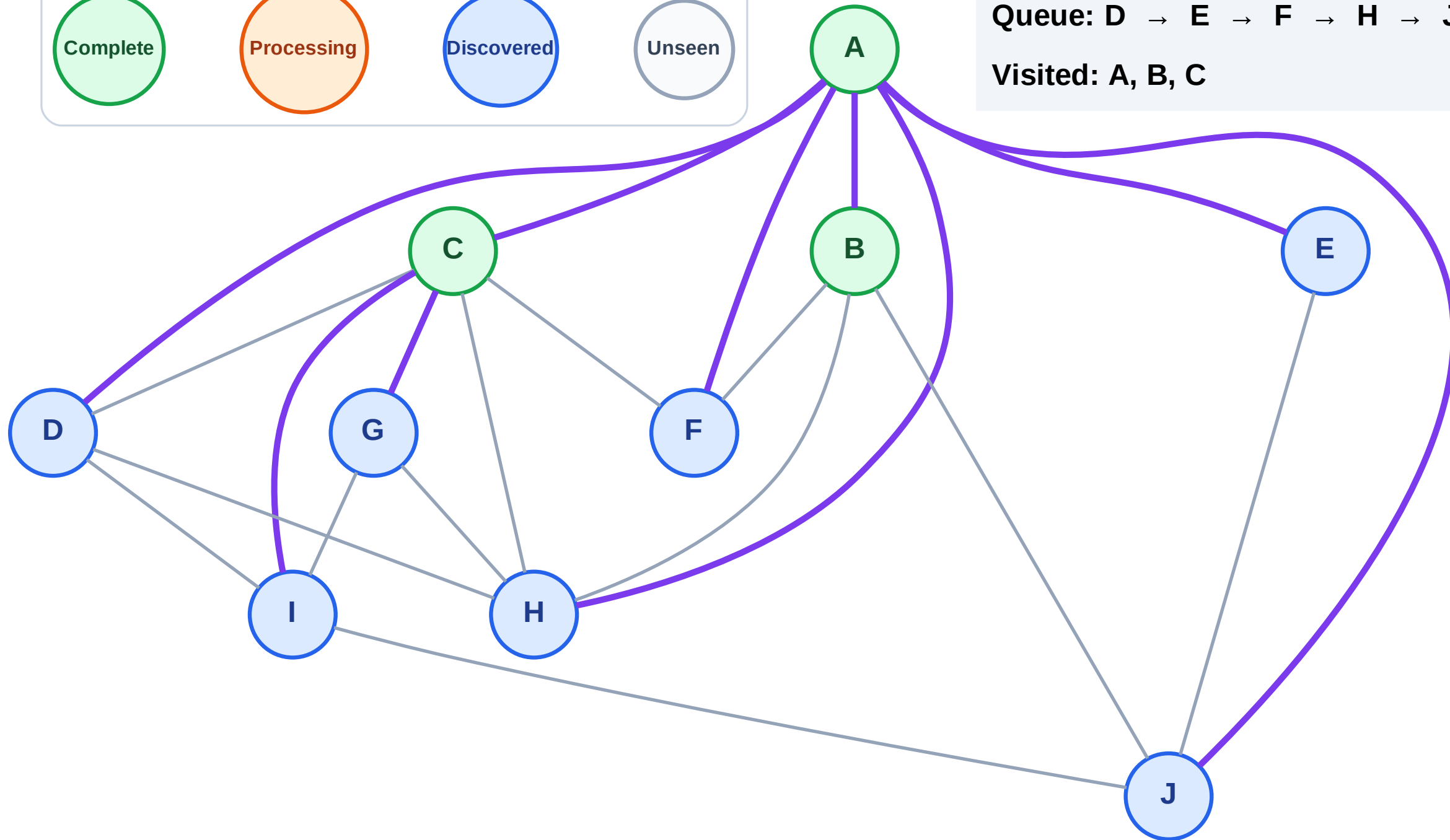
Step 7: Discover G, I from C

Legend



Queue: D → E → F → H → J → G → I

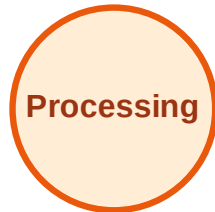
Visited: A, B, C



BFS Traversal

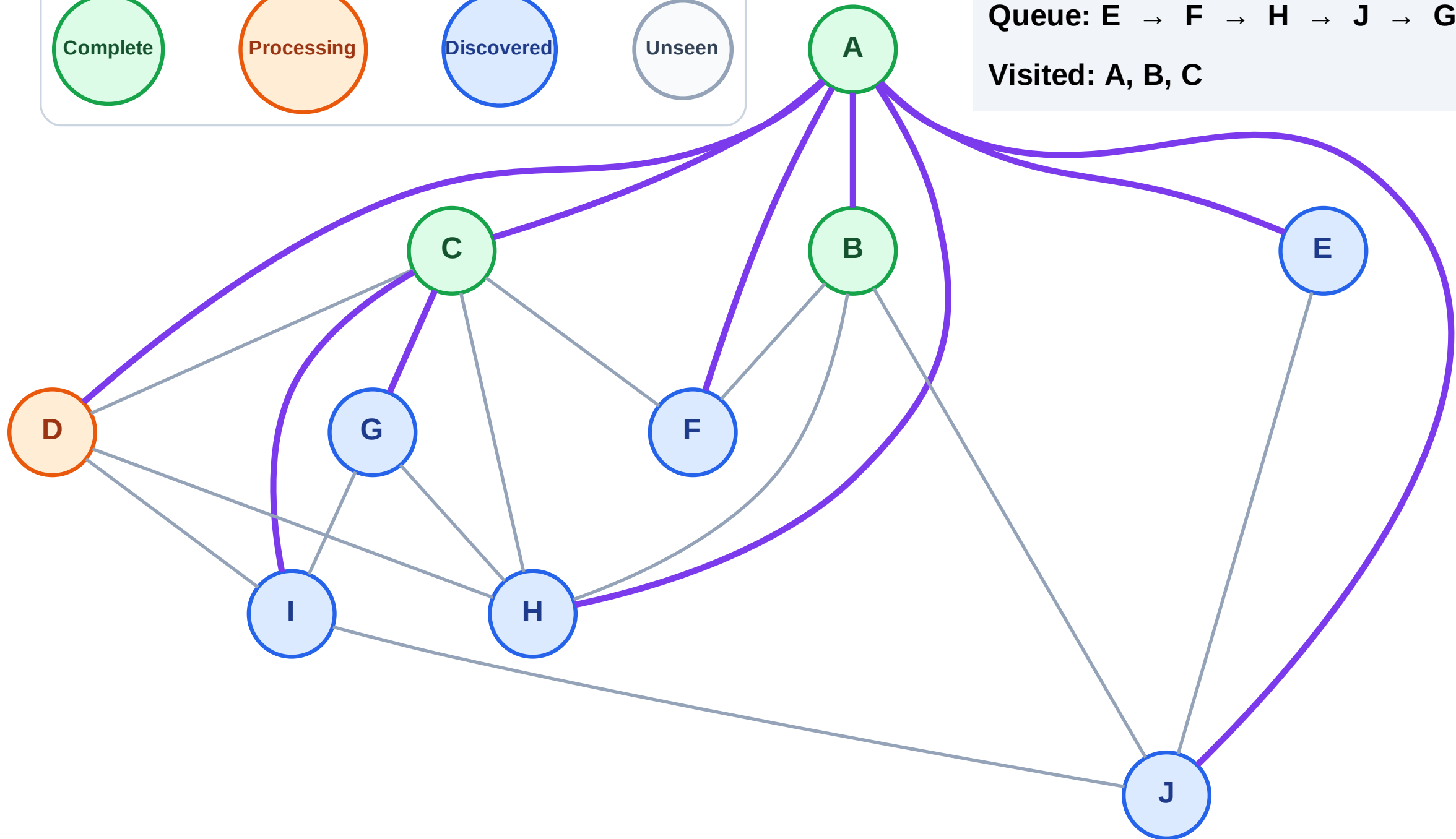
Step 8: Process node D

Legend



Queue: E → F → H → J → G → I

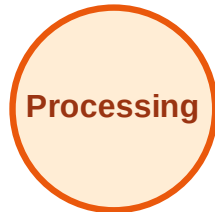
Visited: A, B, C



BFS Traversal

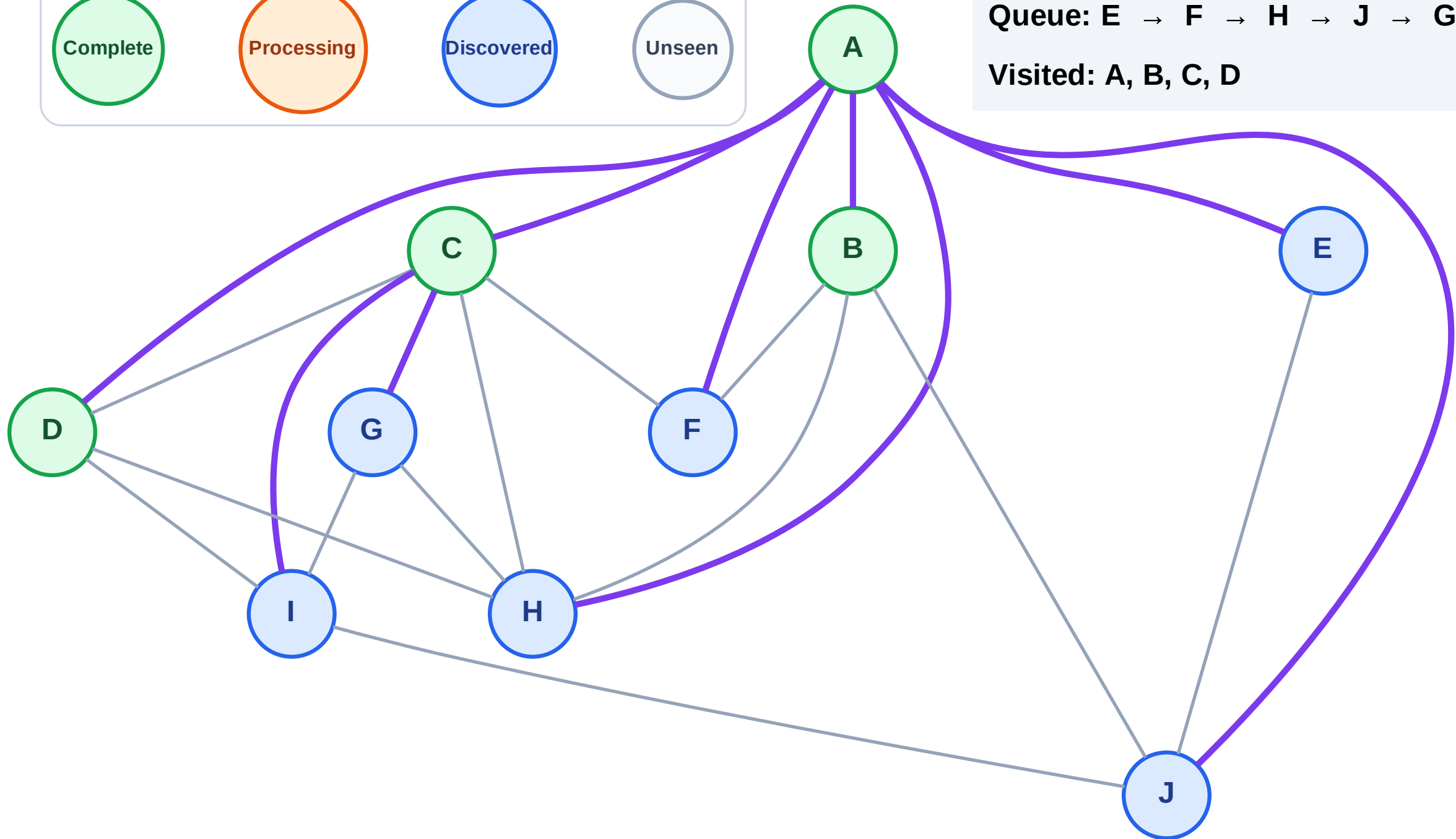
Step 9: No new neighbors from D

Legend



Queue: E → F → H → J → G → I

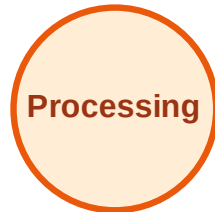
Visited: A, B, C, D



BFS Traversal

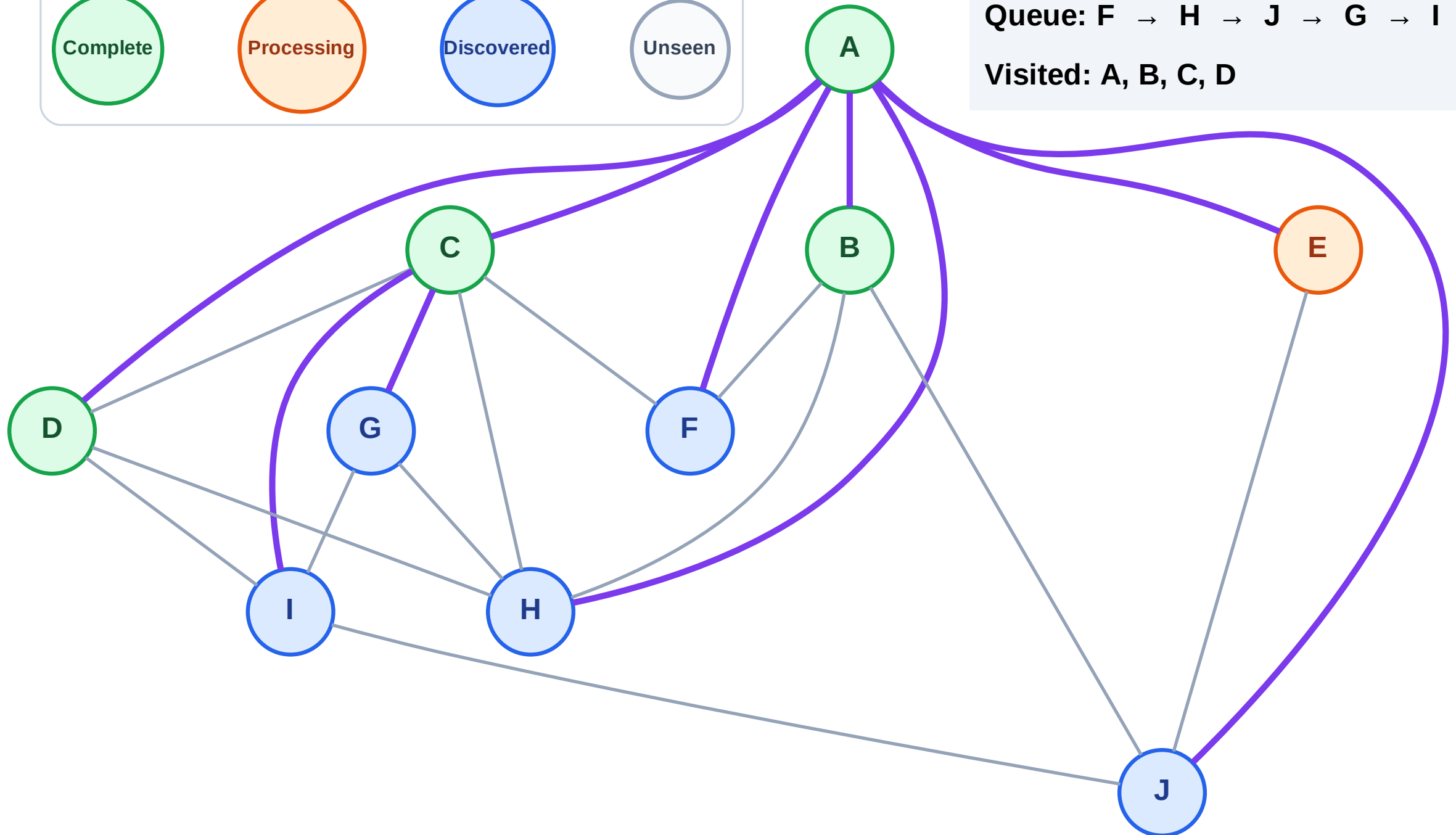
Step 10: Process node E

Legend



Queue: F → H → J → G → I

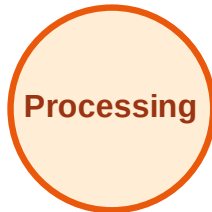
Visited: A, B, C, D



BFS Traversal

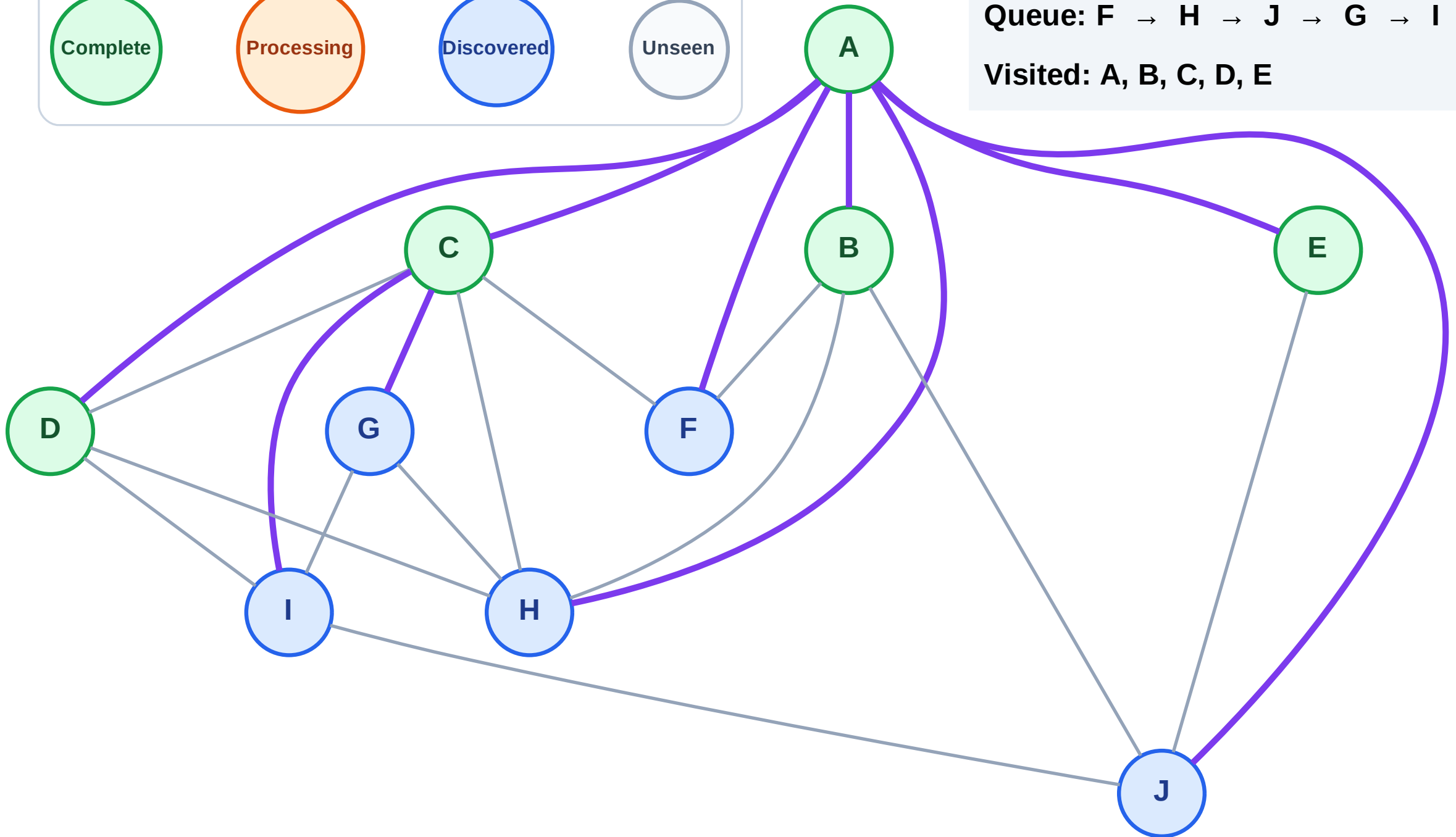
Step 11: No new neighbors from E

Legend



Queue: F → H → J → G → I

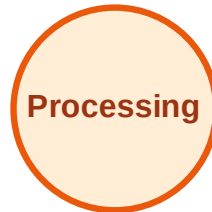
Visited: A, B, C, D, E



BFS Traversal

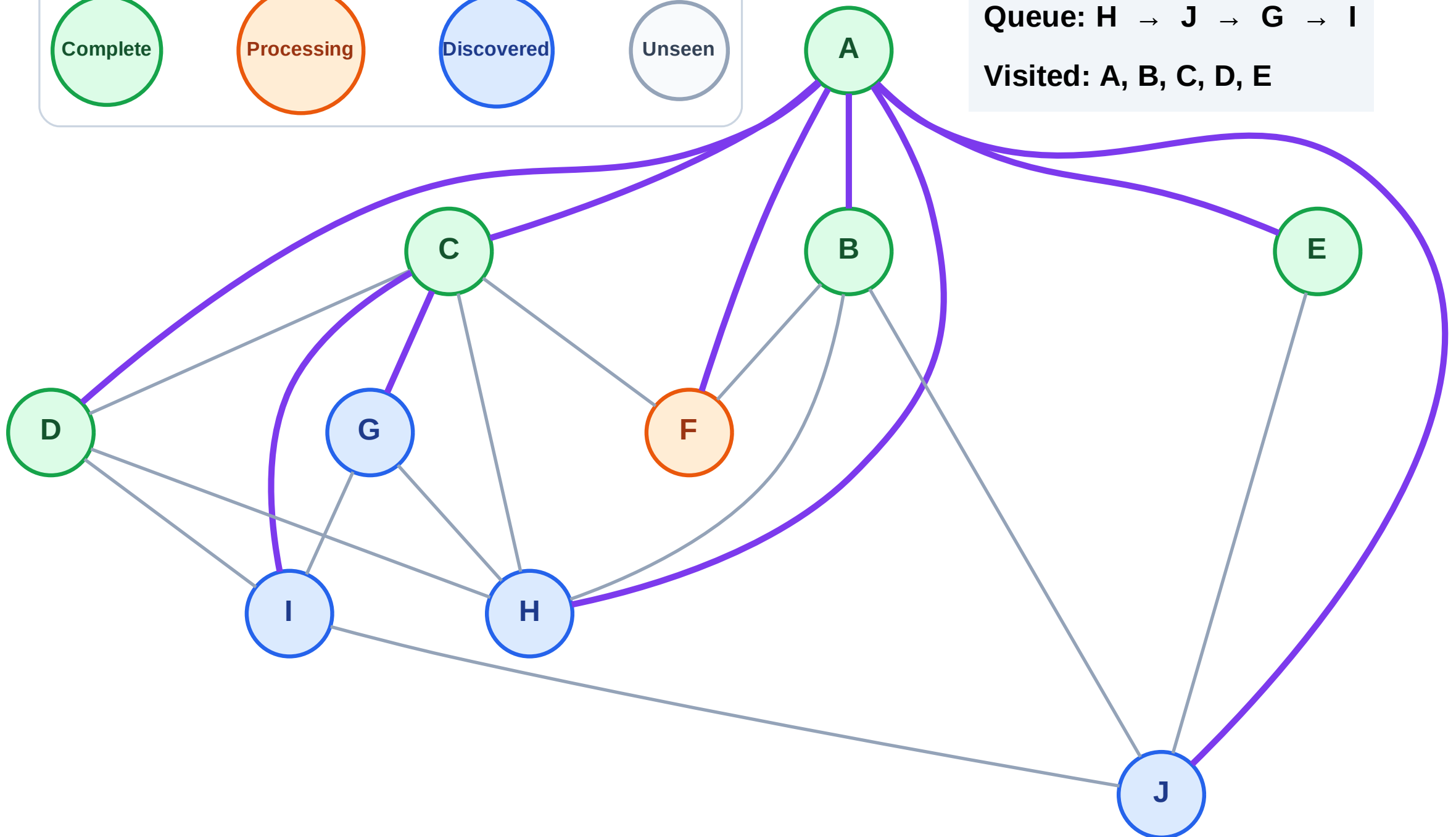
Step 12: Process node F

Legend



Queue: H → J → G → I

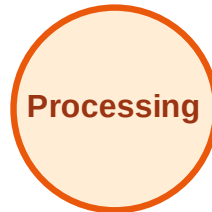
Visited: A, B, C, D, E



BFS Traversal

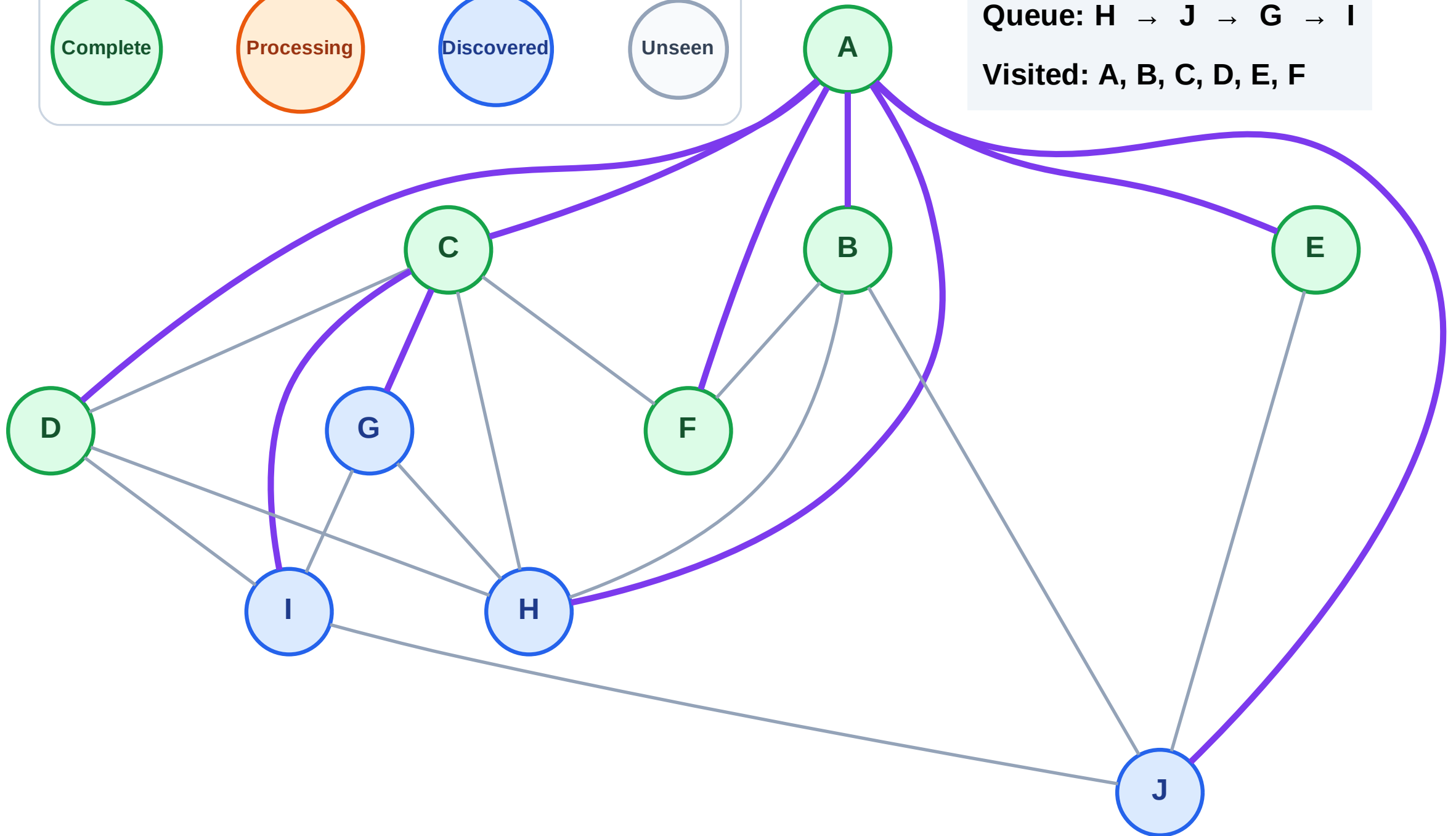
Step 13: No new neighbors from F

Legend



Queue: H → J → G → I

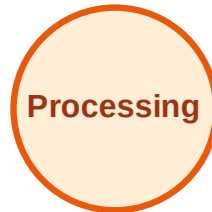
Visited: A, B, C, D, E, F



BFS Traversal

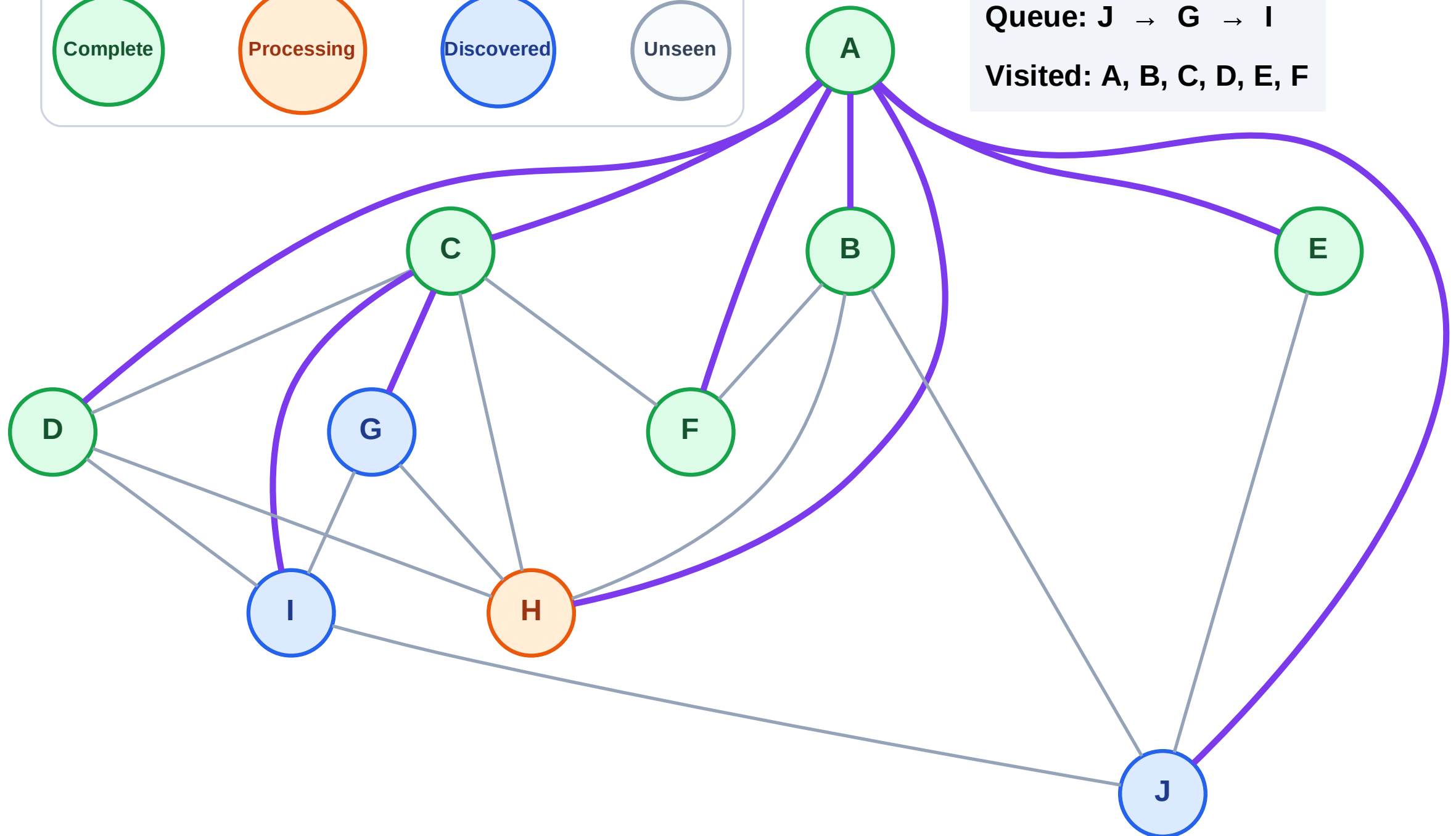
Step 14: Process node H

Legend



Queue: J → G → I

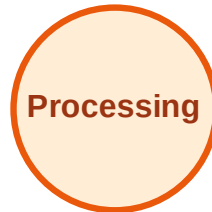
Visited: A, B, C, D, E, F



BFS Traversal

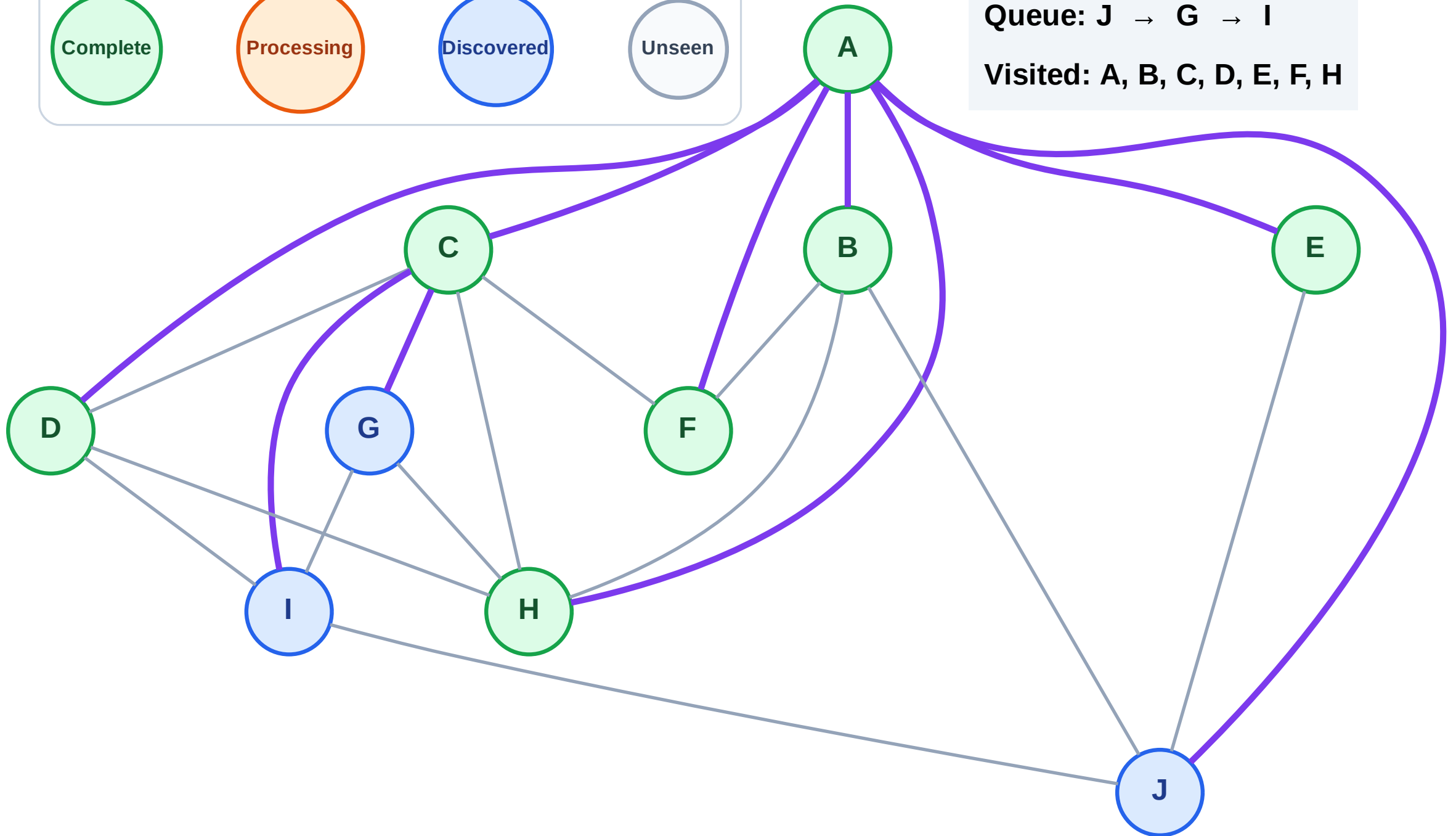
Step 15: No new neighbors from H

Legend



Queue: J → G → I

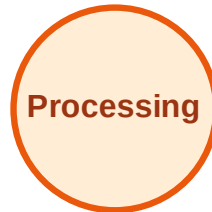
Visited: A, B, C, D, E, F, H



BFS Traversal

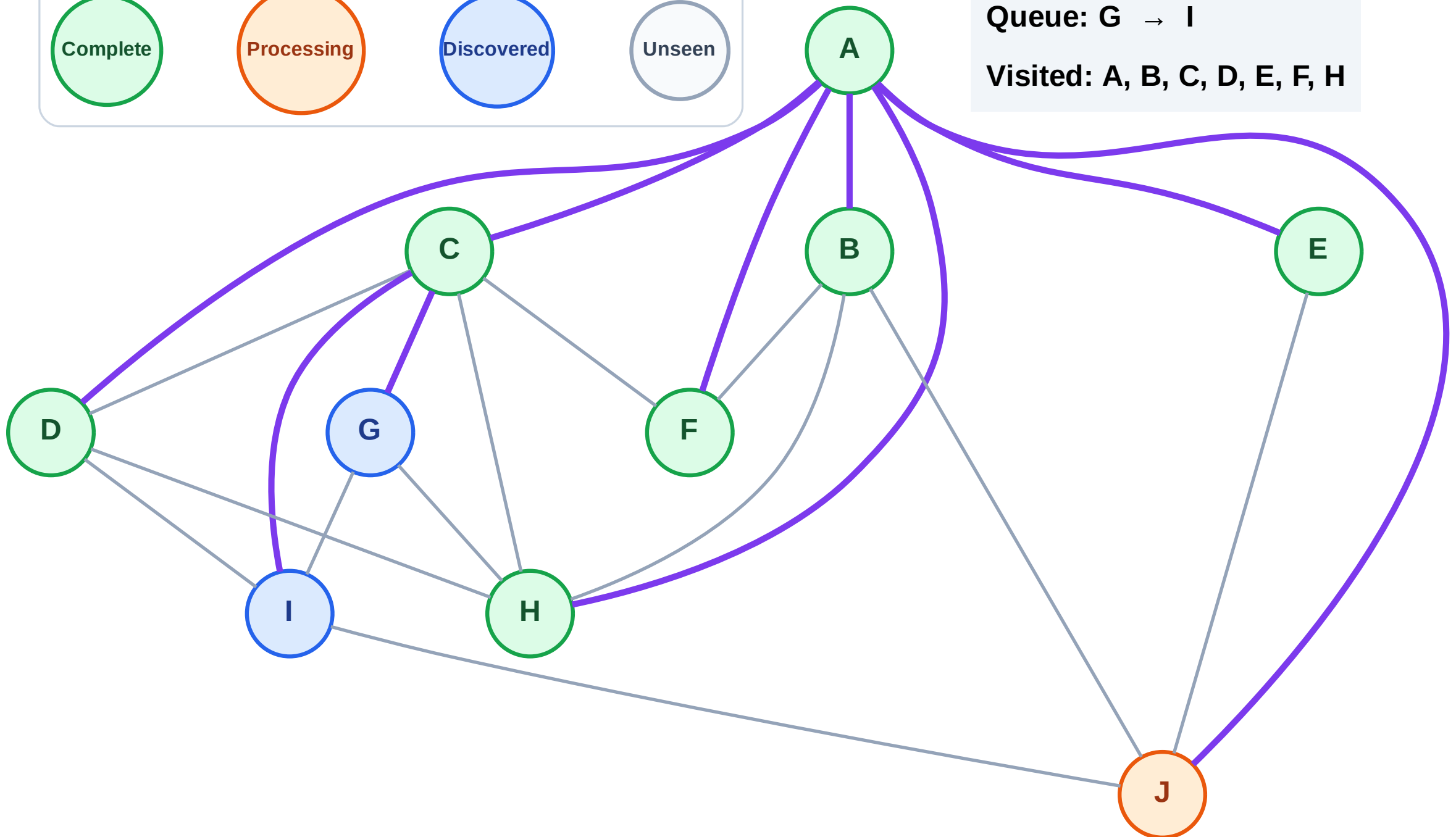
Step 16: Process node J

Legend



Queue: G → I

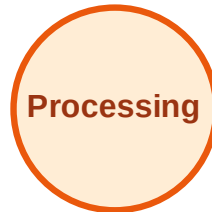
Visited: A, B, C, D, E, F, H



BFS Traversal

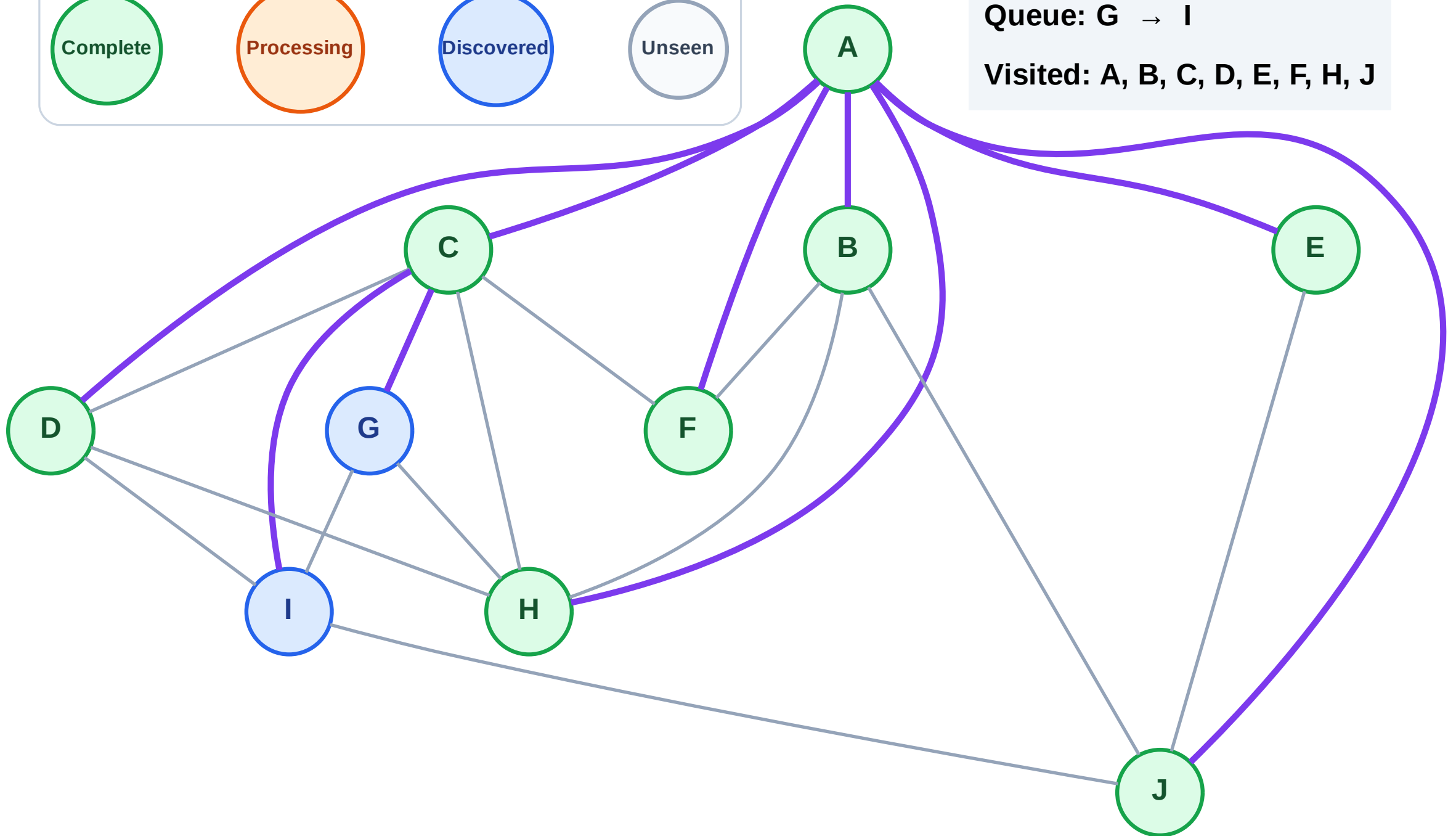
Step 17: No new neighbors from J

Legend



Queue: G → I

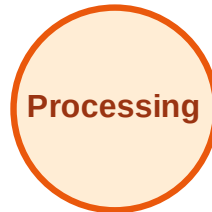
Visited: A, B, C, D, E, F, H, J



BFS Traversal

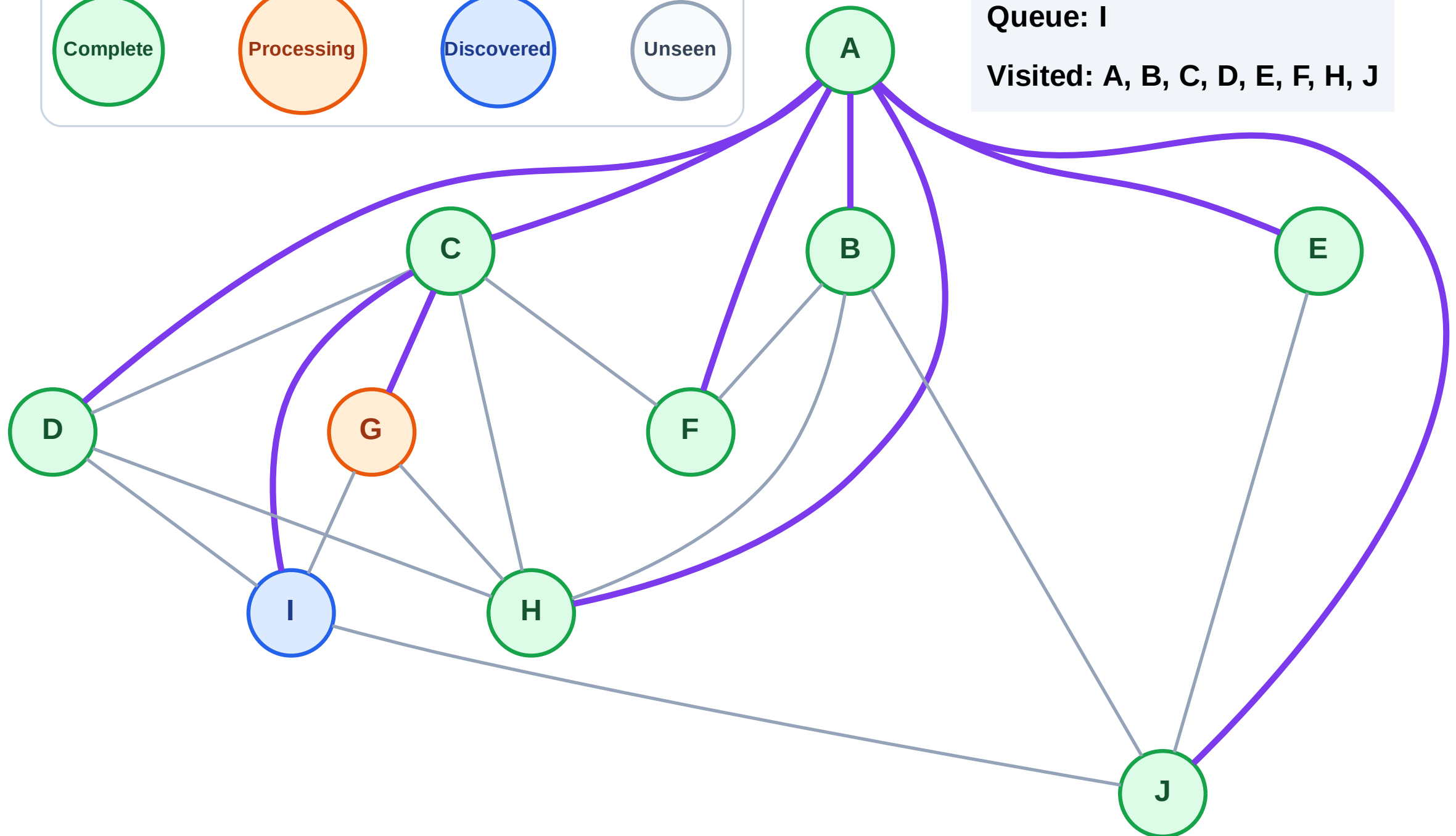
Step 18: Process node G

Legend



Queue: I

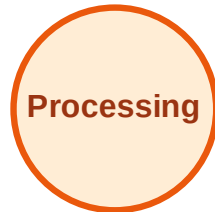
Visited: A, B, C, D, E, F, H, J



BFS Traversal

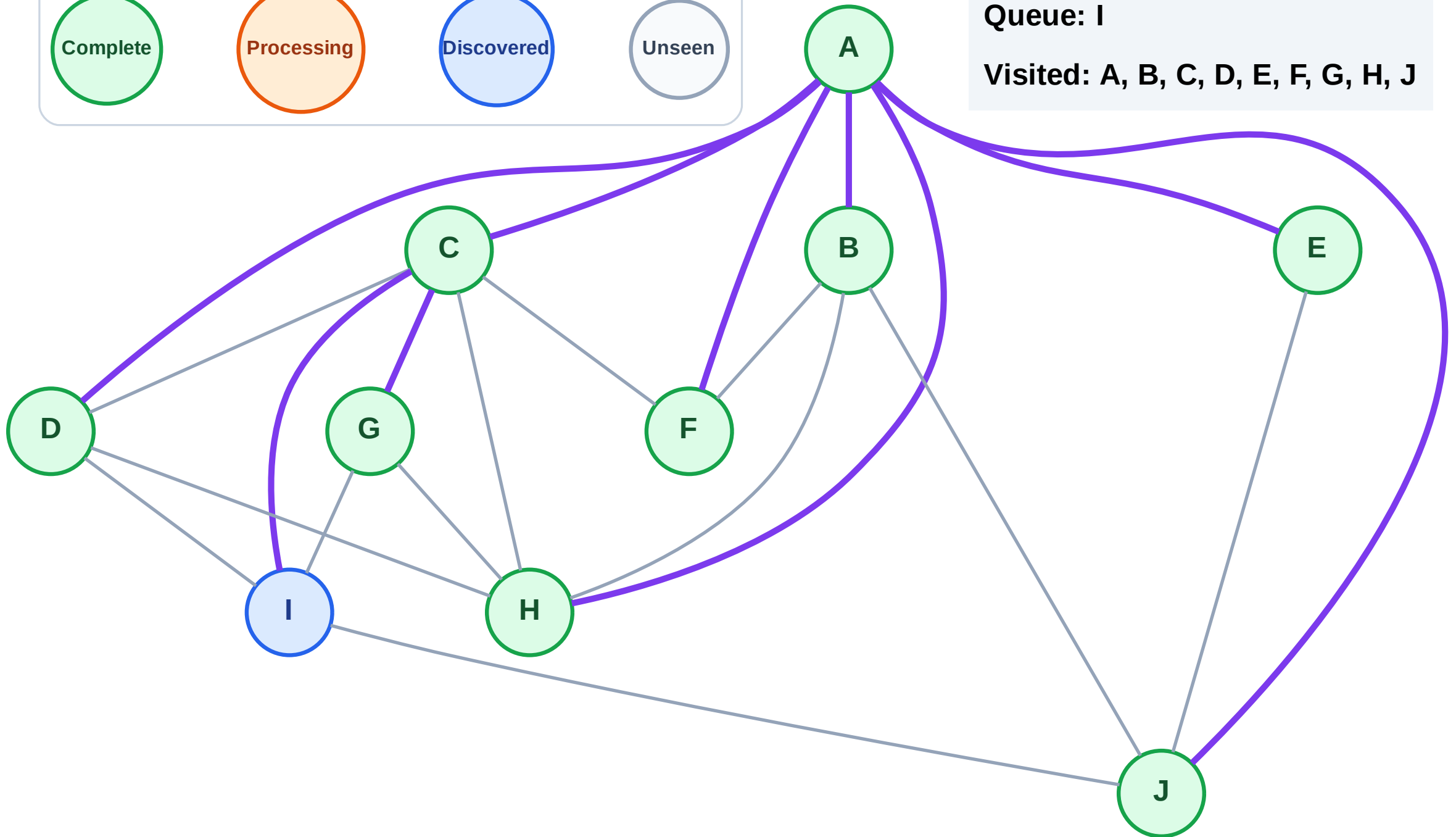
Step 19: No new neighbors from G

Legend



Queue: I

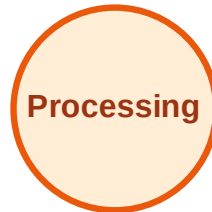
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

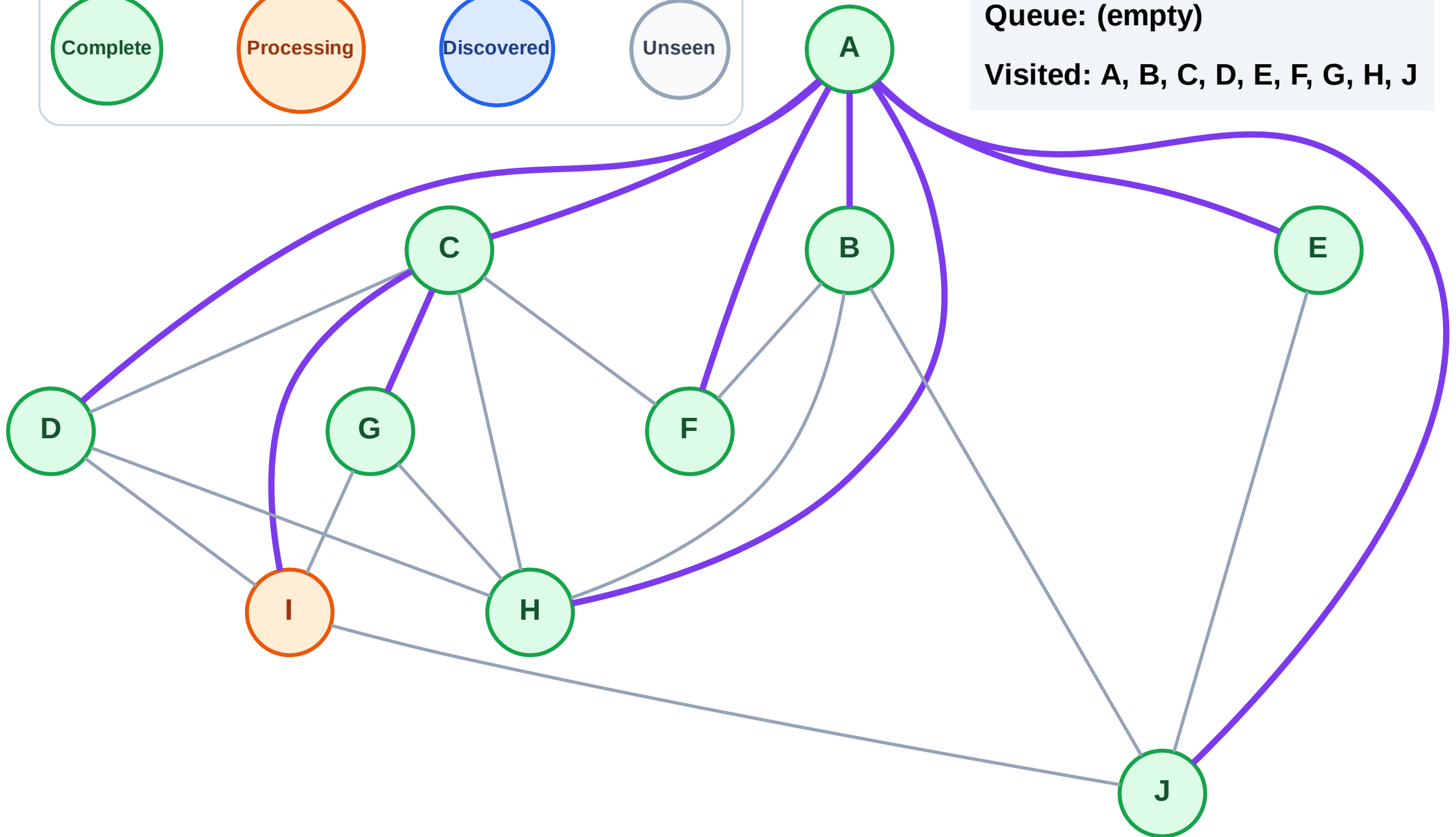
Step 20: Process node I

Legend



Queue: (empty)

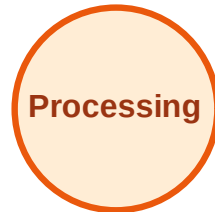
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

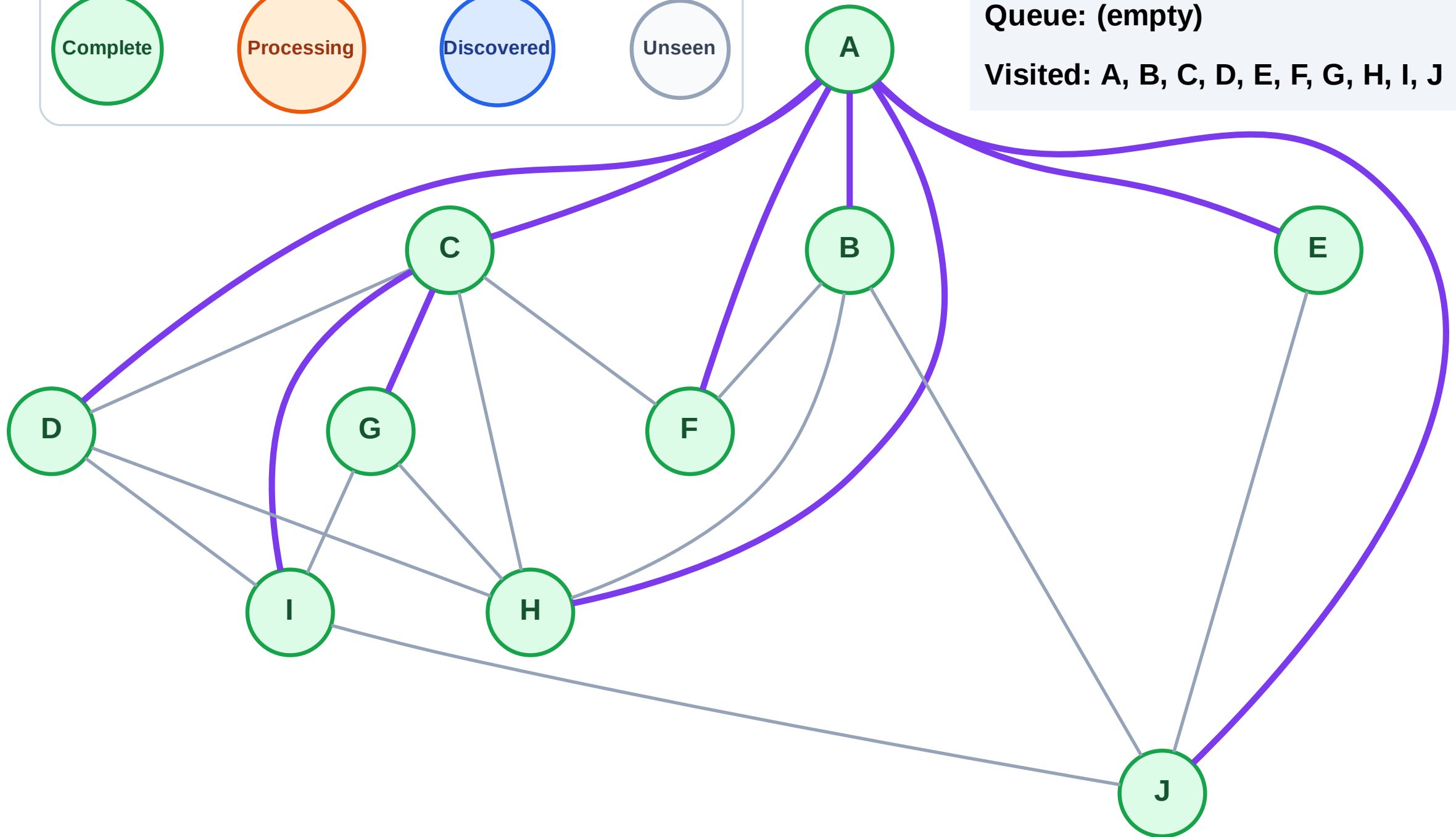
Step 21: No new neighbors from I

Legend



Queue: (empty)

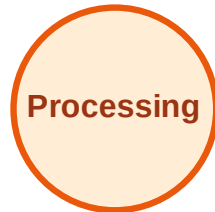
Visited: A, B, C, D, E, F, G, H, I, J



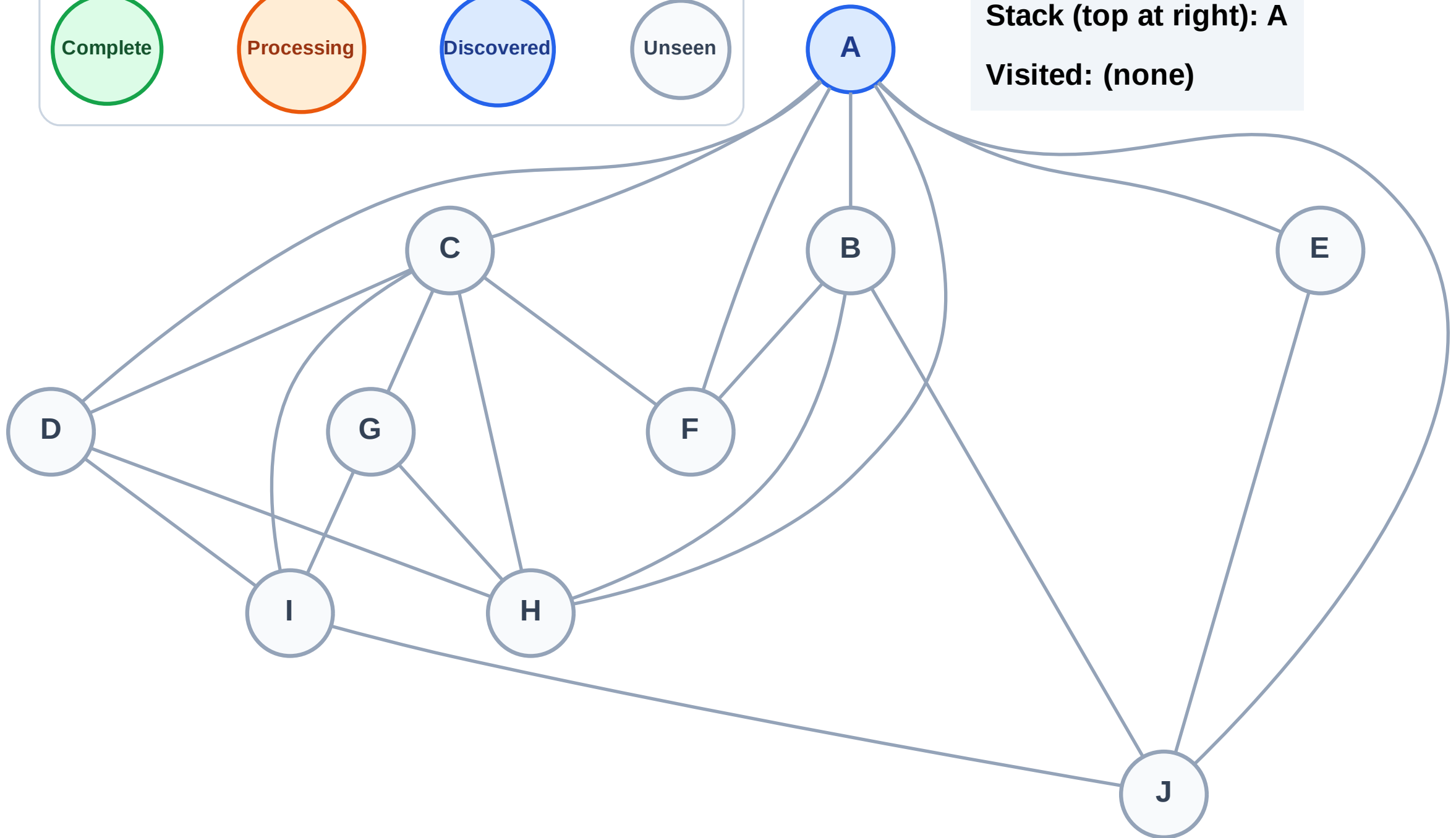
DFS Traversal

Step 1: Start at A

Legend



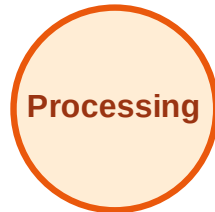
Stack (top at right): A
Visited: (none)



DFS Traversal

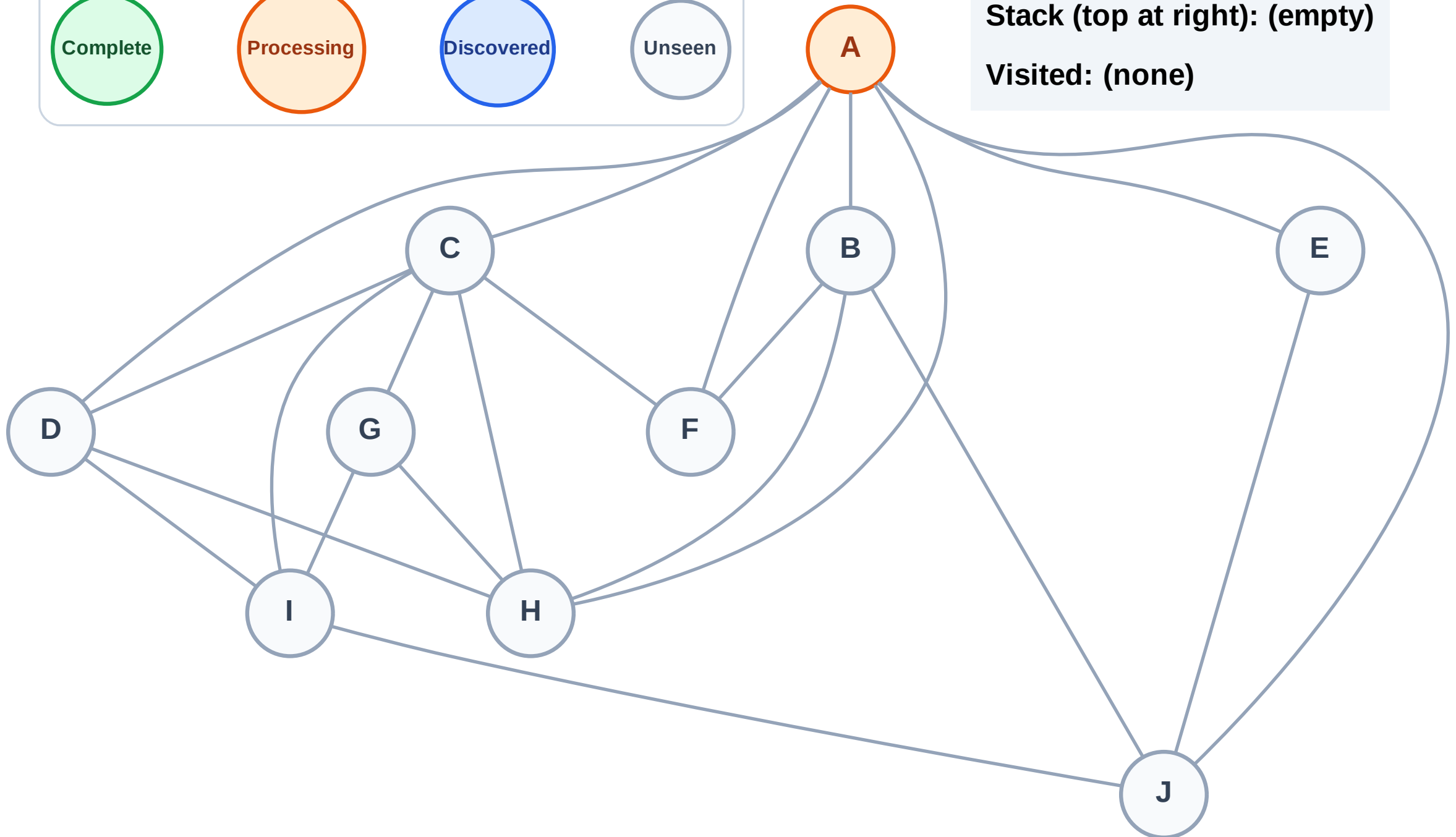
Step 2: Process node A

Legend



Stack (top at right): (empty)

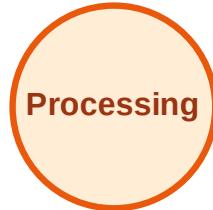
Visited: (none)



DFS Traversal

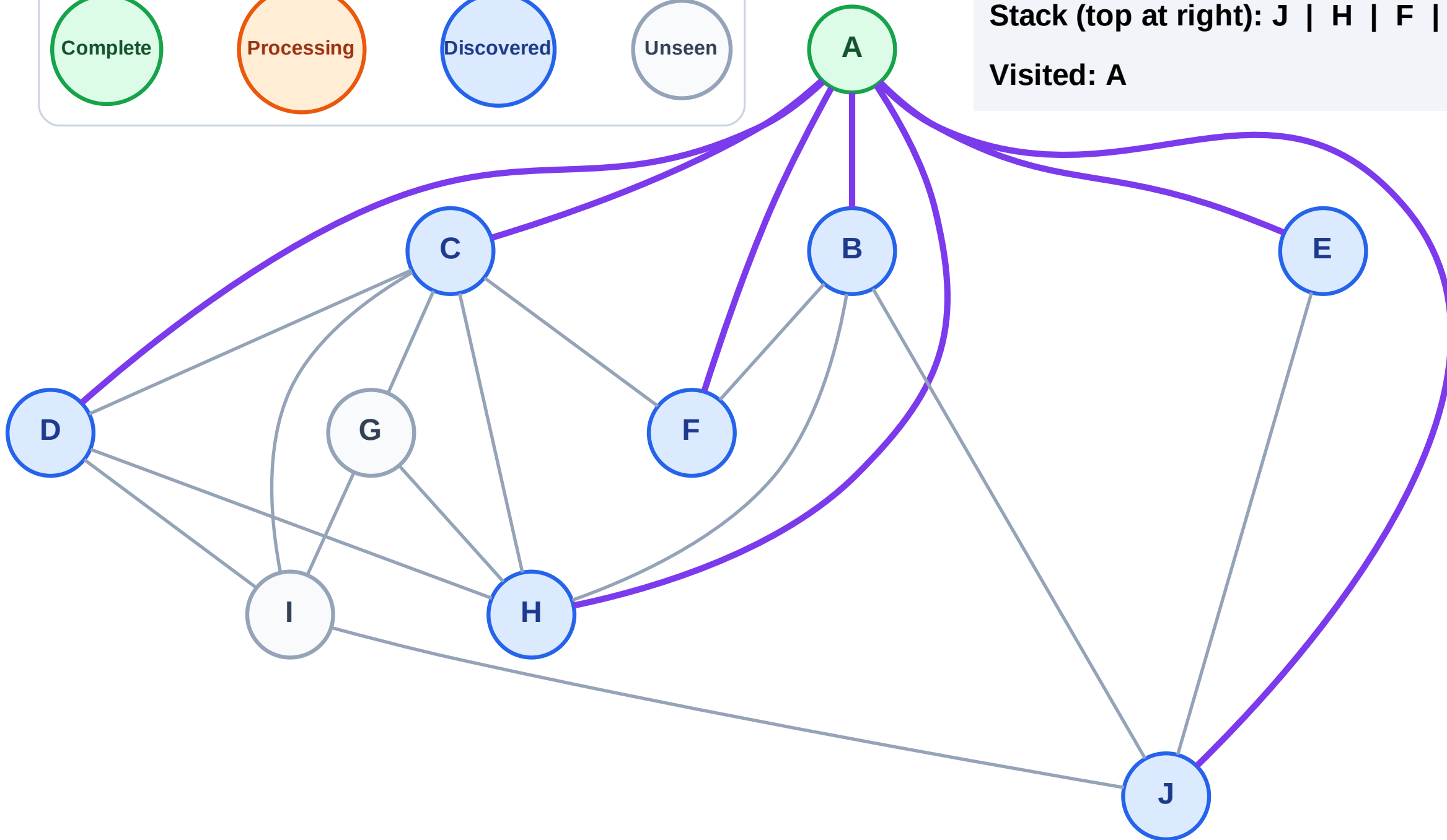
Step 3: Discover J, H, F, E, D, C, B from A

Legend



Stack (top at right): J | H | F | E | D | C | B

Visited: A



DFS Traversal

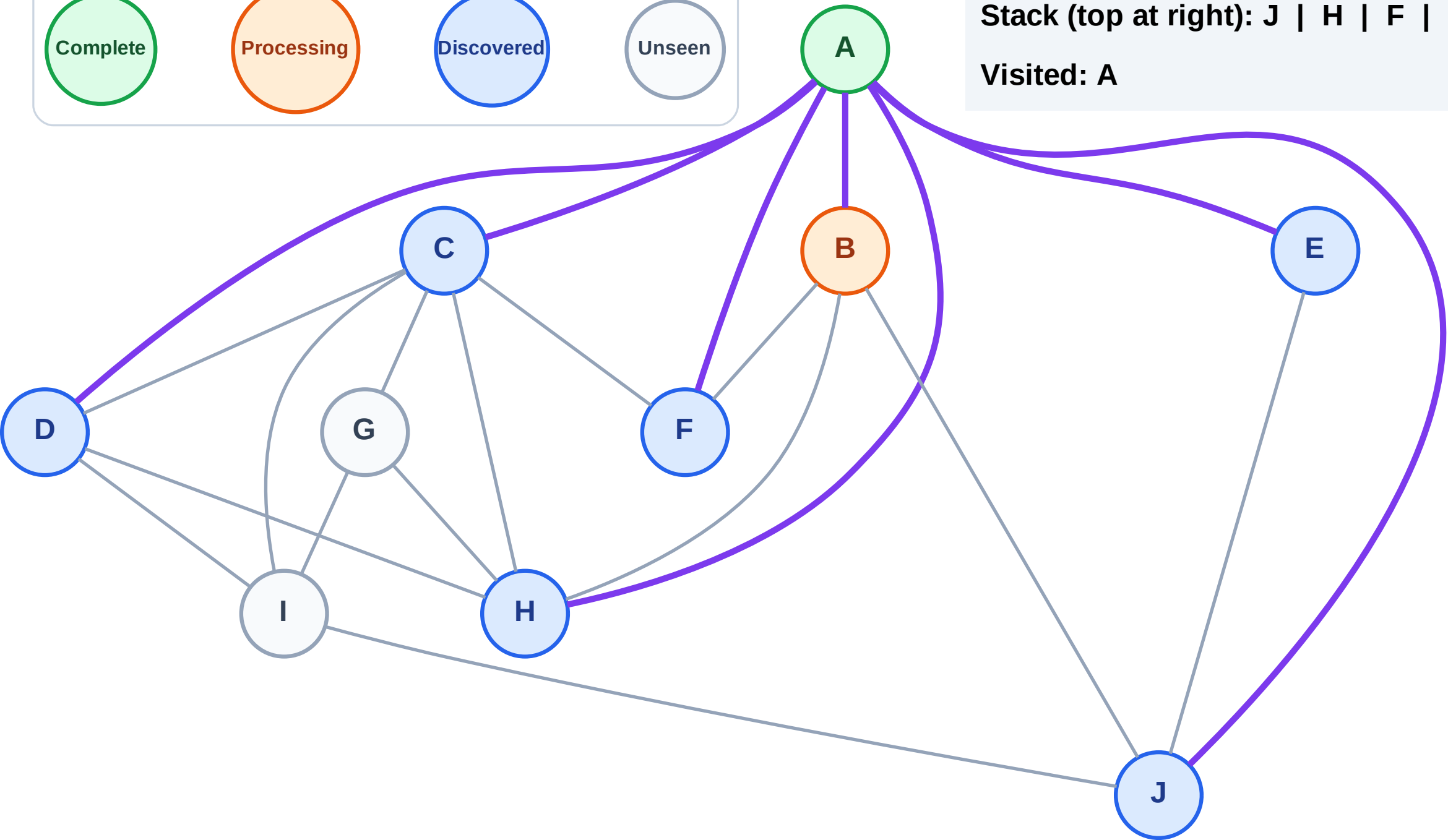
Step 4: Process node B

Legend



Stack (top at right): J | H | F | E | D | C

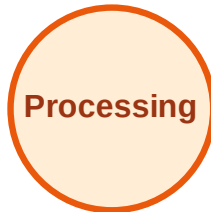
Visited: A



DFS Traversal

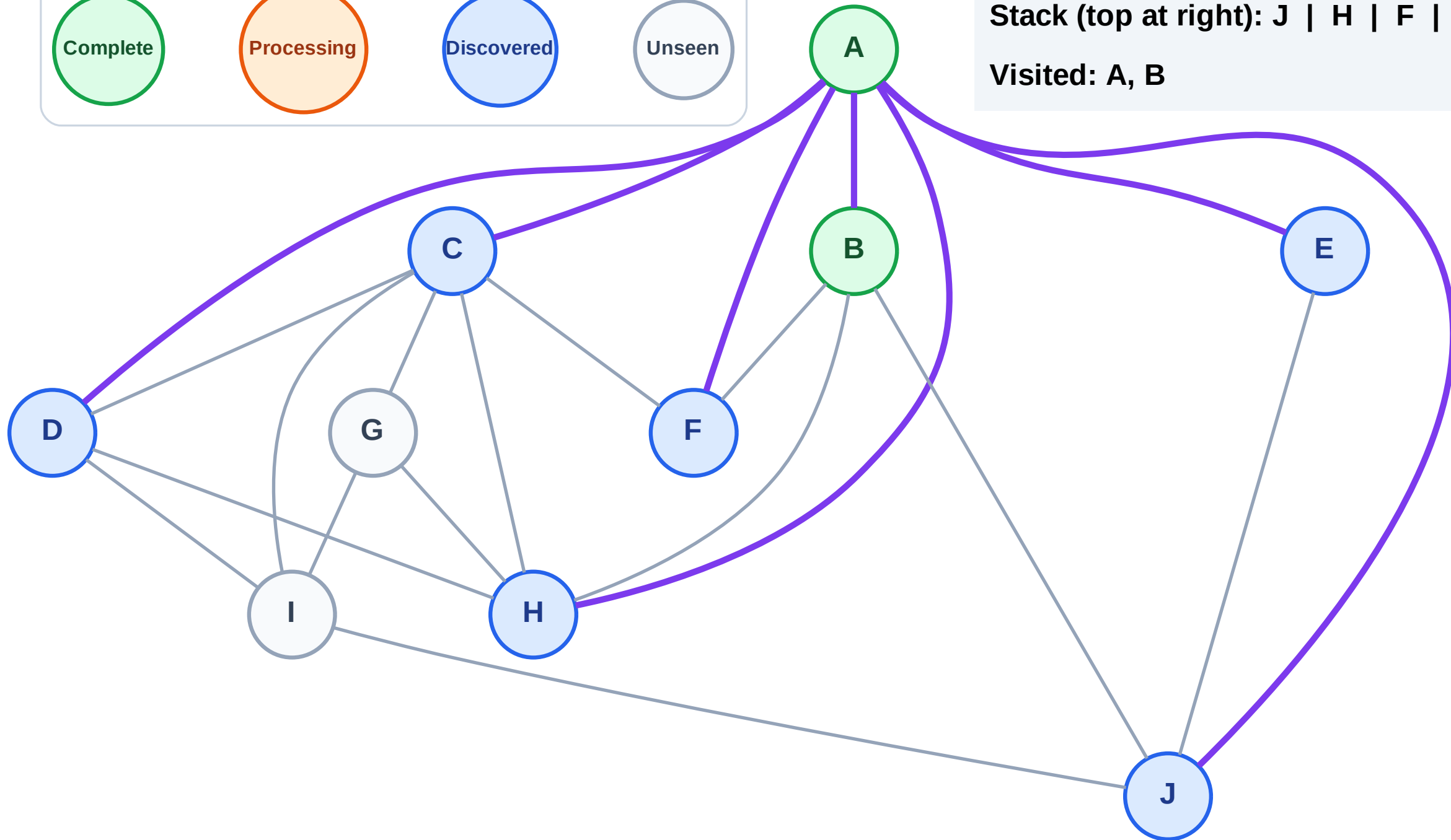
Step 5: No new neighbors from B

Legend



Stack (top at right): J | H | F | E | D | C

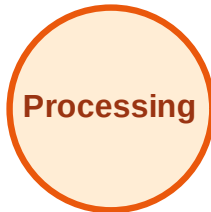
Visited: A, B



DFS Traversal

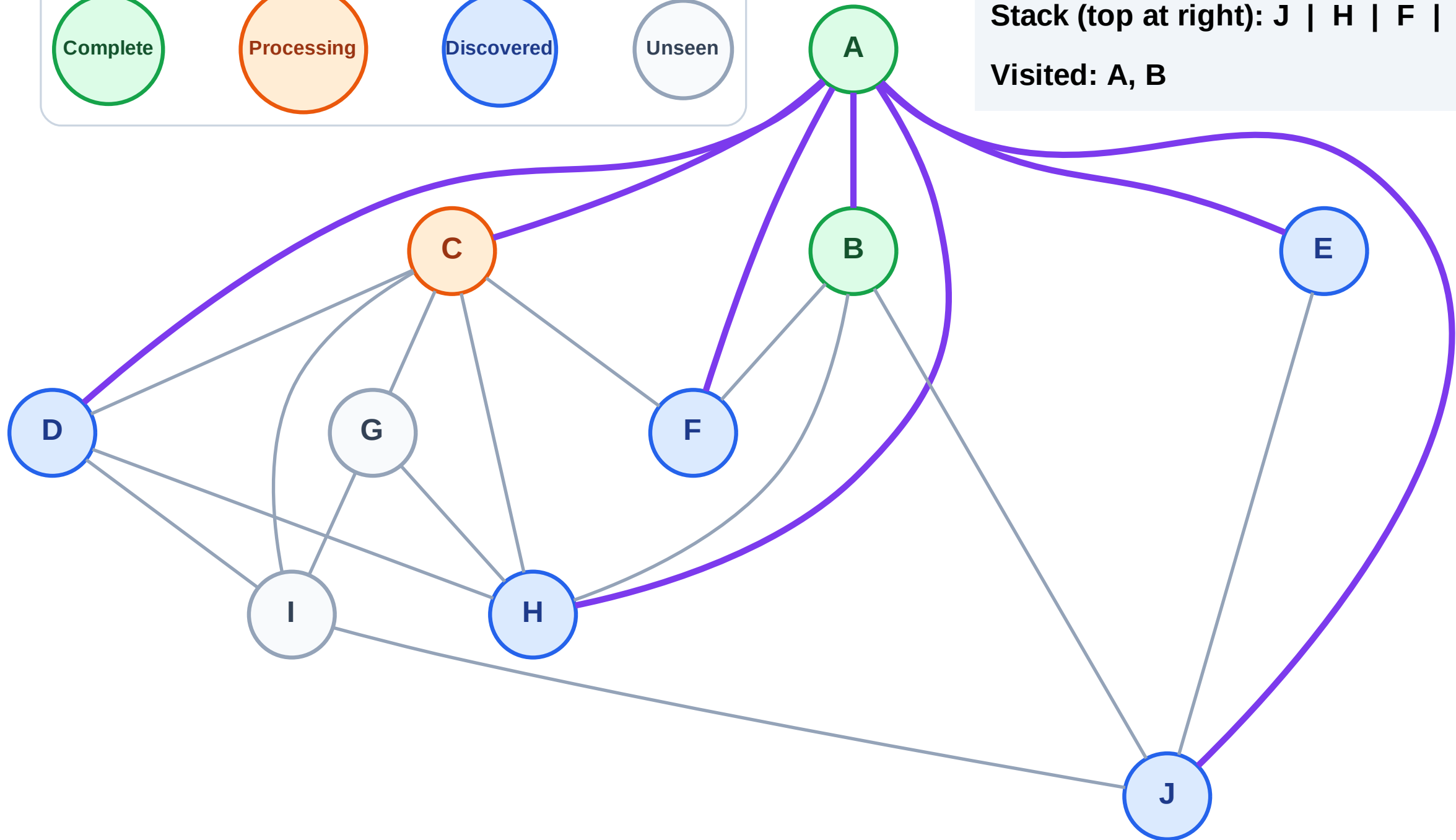
Step 6: Process node C

Legend



Stack (top at right): J | H | F | E | D

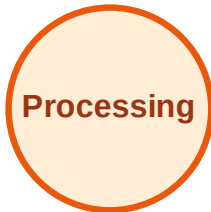
Visited: A, B



DFS Traversal

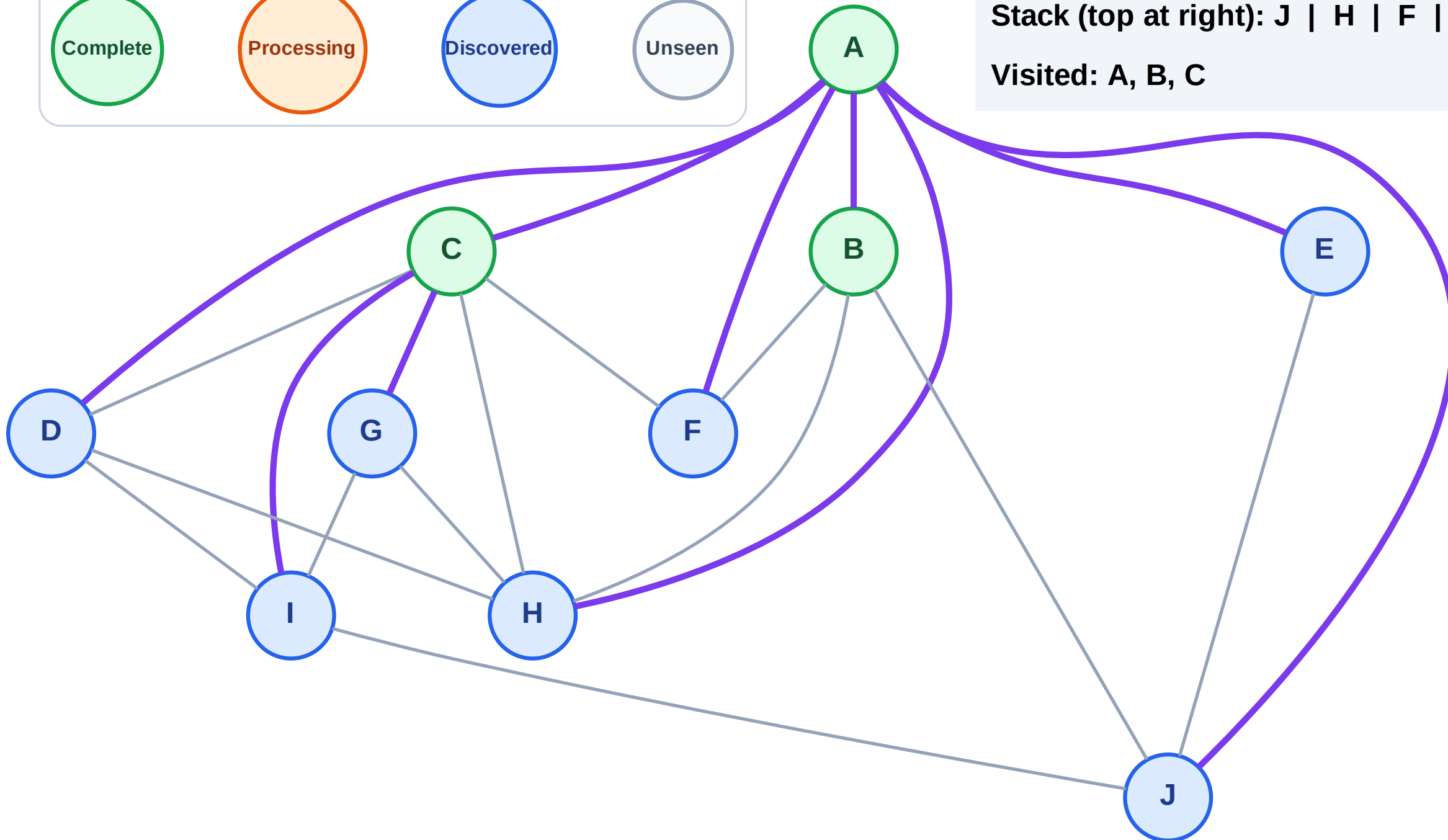
Step 7: Discover I, G from C

Legend



Stack (top at right): J | H | F | E | D | I | G

Visited: A, B, C



DFS Traversal

Step 8: Process node G

Legend

Complete

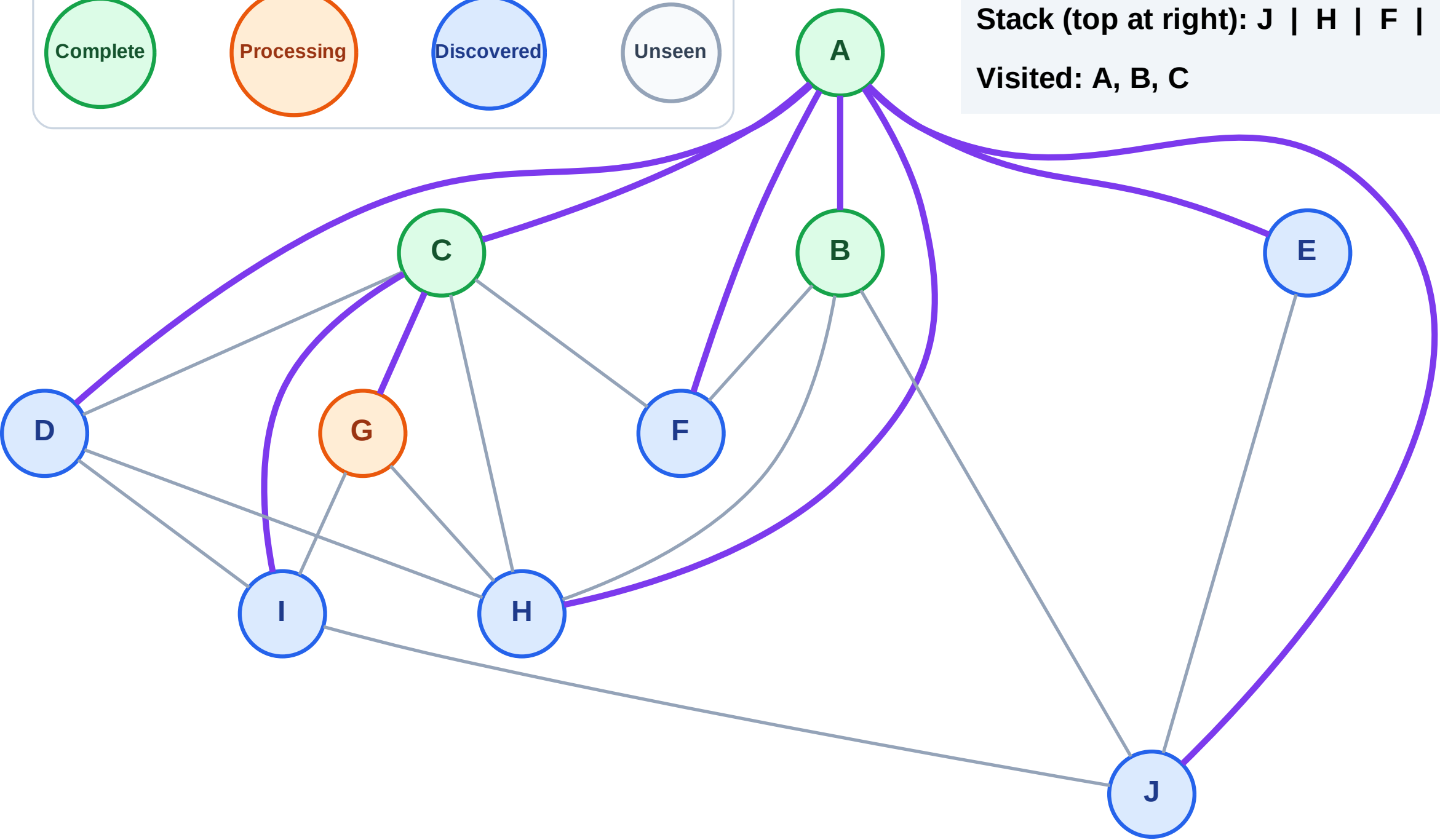
Processing

Discovered

Unseen

Stack (top at right): J | H | F | E | D | I

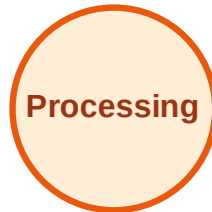
Visited: A, B, C



DFS Traversal

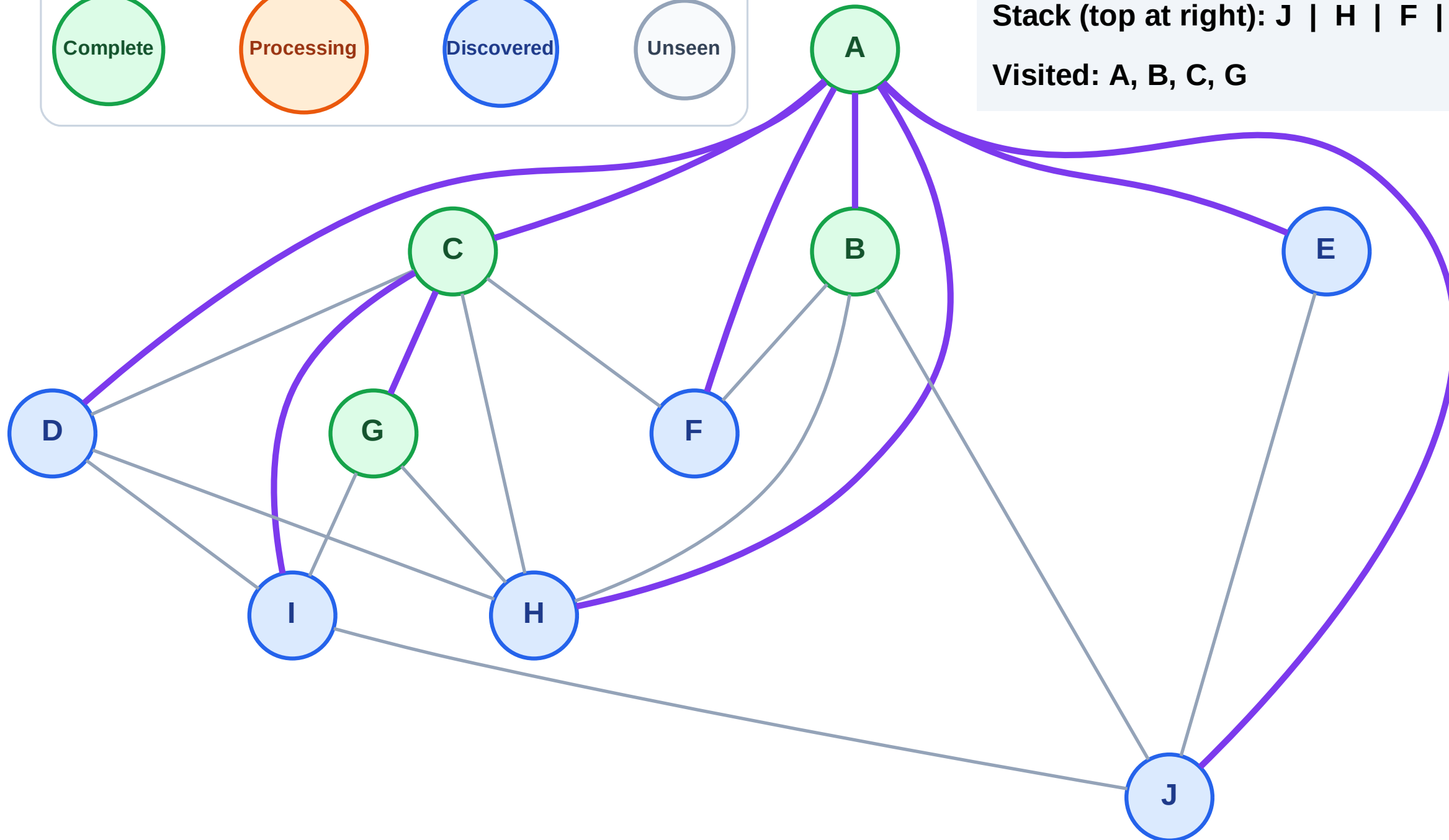
Step 9: No new neighbors from G

Legend



Stack (top at right): J | H | F | E | D | I

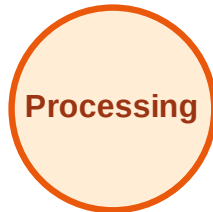
Visited: A, B, C, G



DFS Traversal

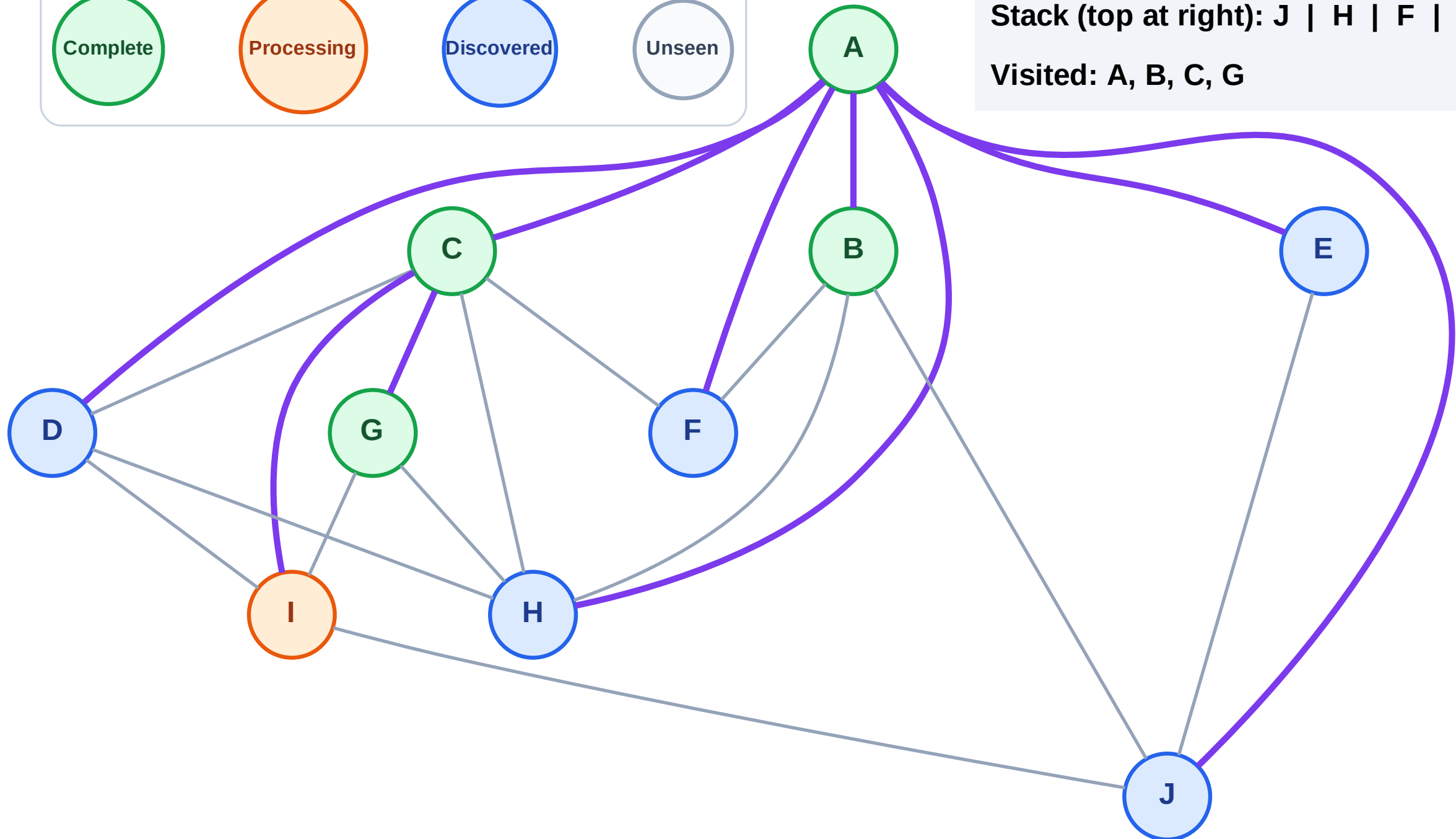
Step 10: Process node I

Legend



Stack (top at right): J | H | F | E | D

Visited: A, B, C, G



DFS Traversal

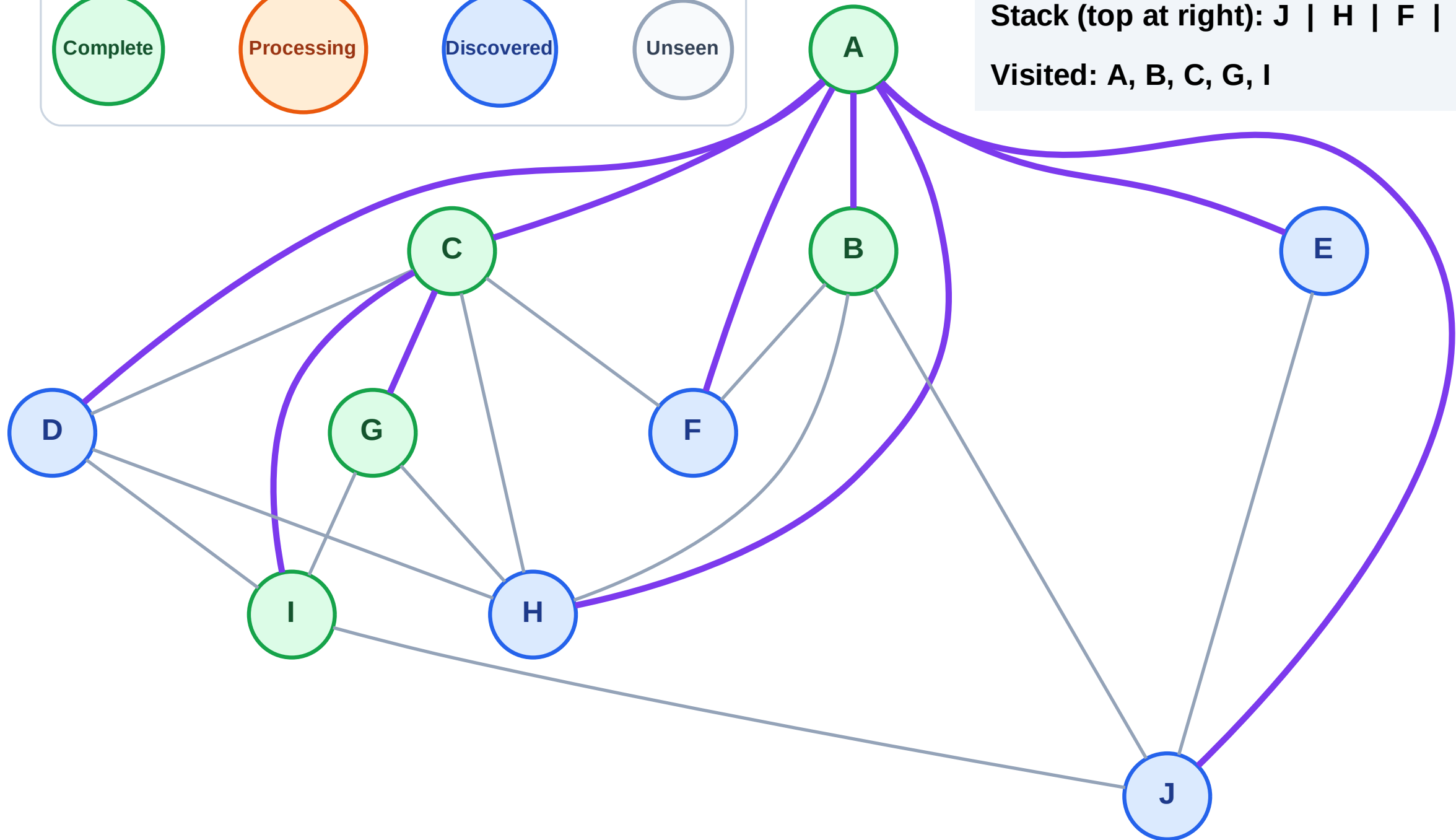
Step 11: No new neighbors from I

Legend



Stack (top at right): J | H | F | E | D

Visited: A, B, C, G, I



DFS Traversal

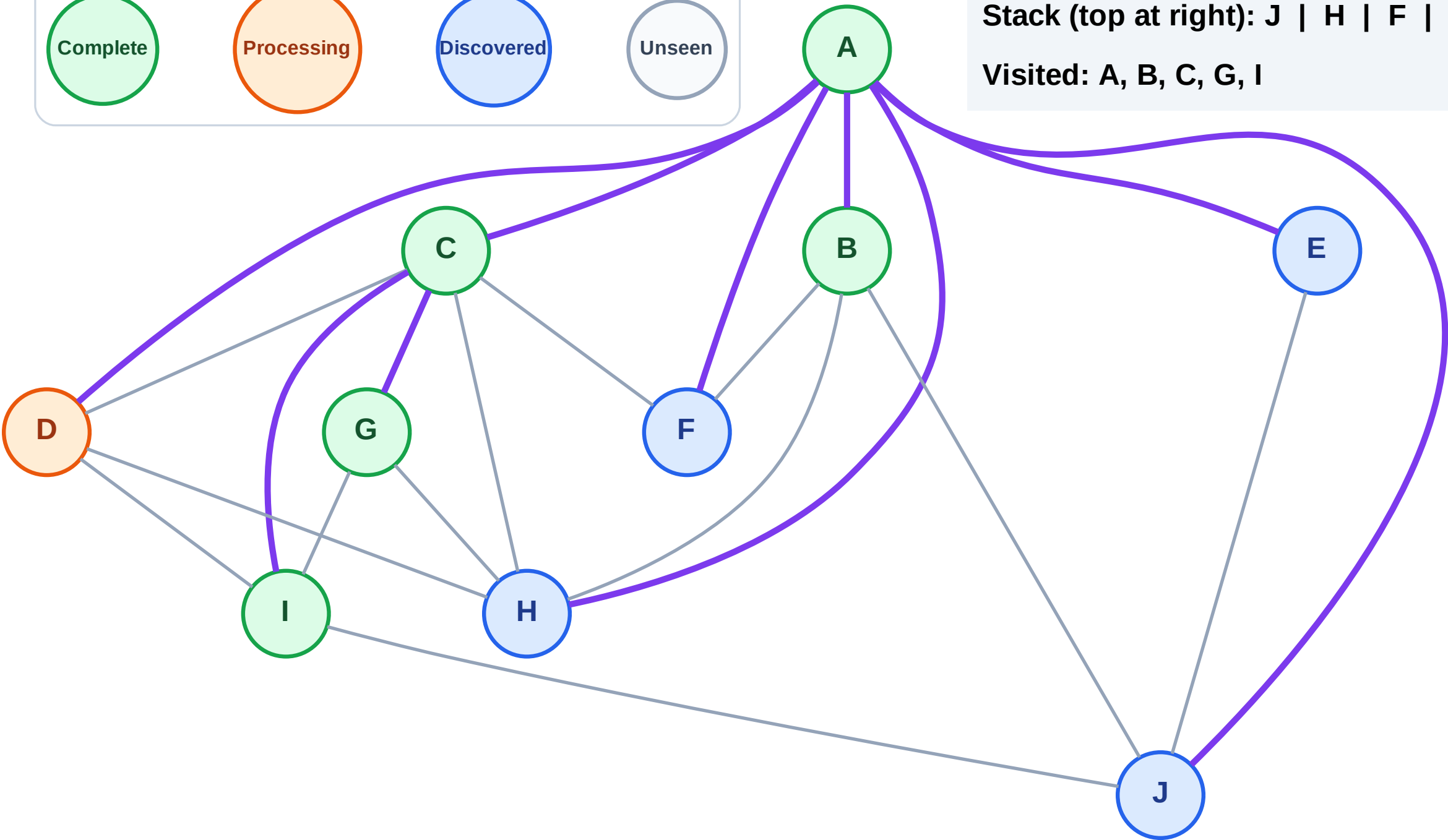
Step 12: Process node D

Legend



Stack (top at right): J | H | F | E

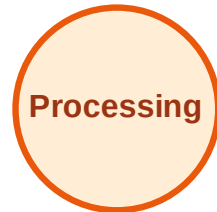
Visited: A, B, C, G, I



DFS Traversal

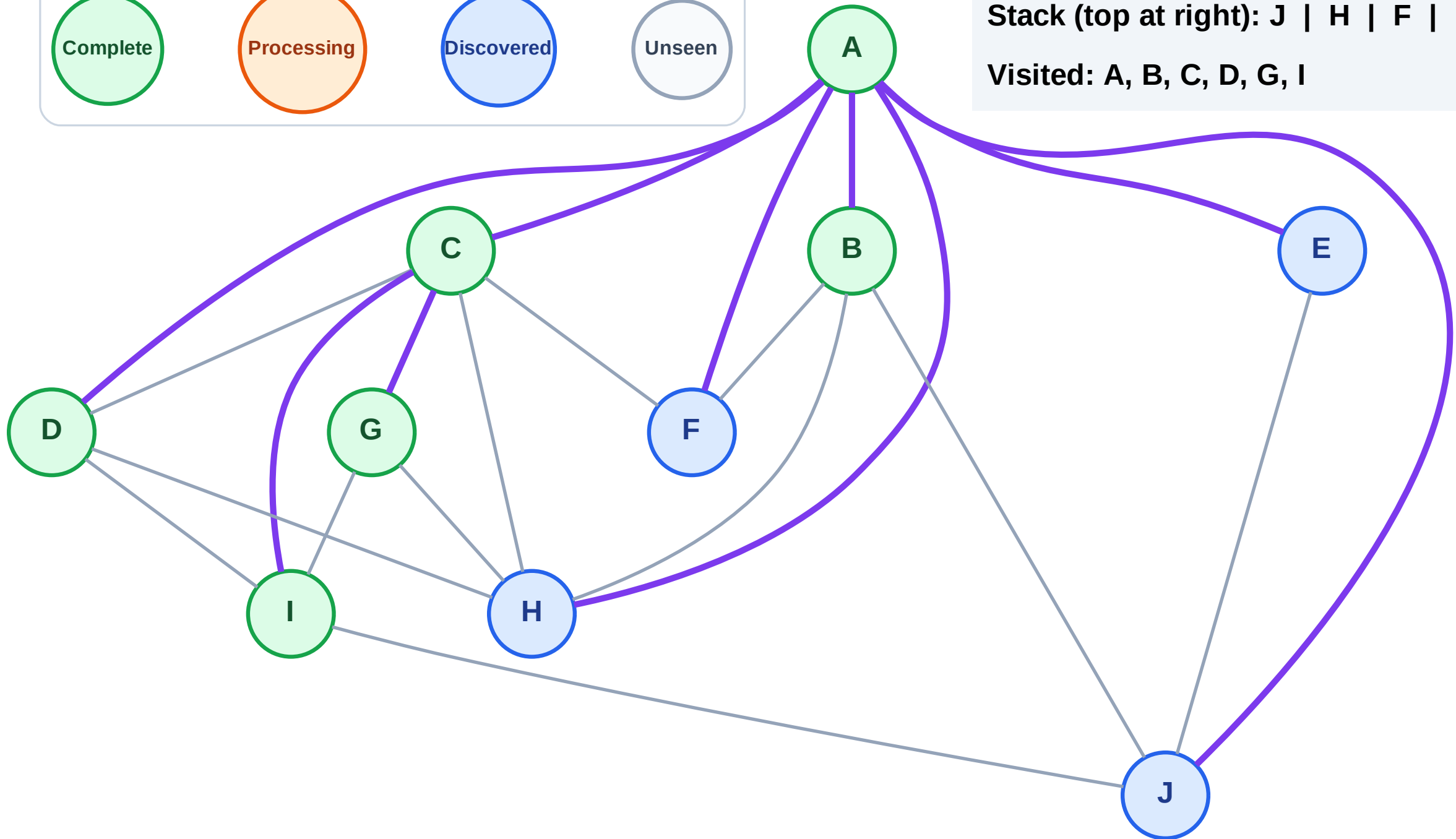
Step 13: No new neighbors from D

Legend



Stack (top at right): J | H | F | E

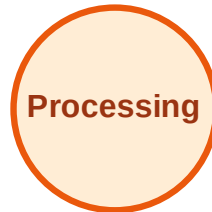
Visited: A, B, C, D, G, I



DFS Traversal

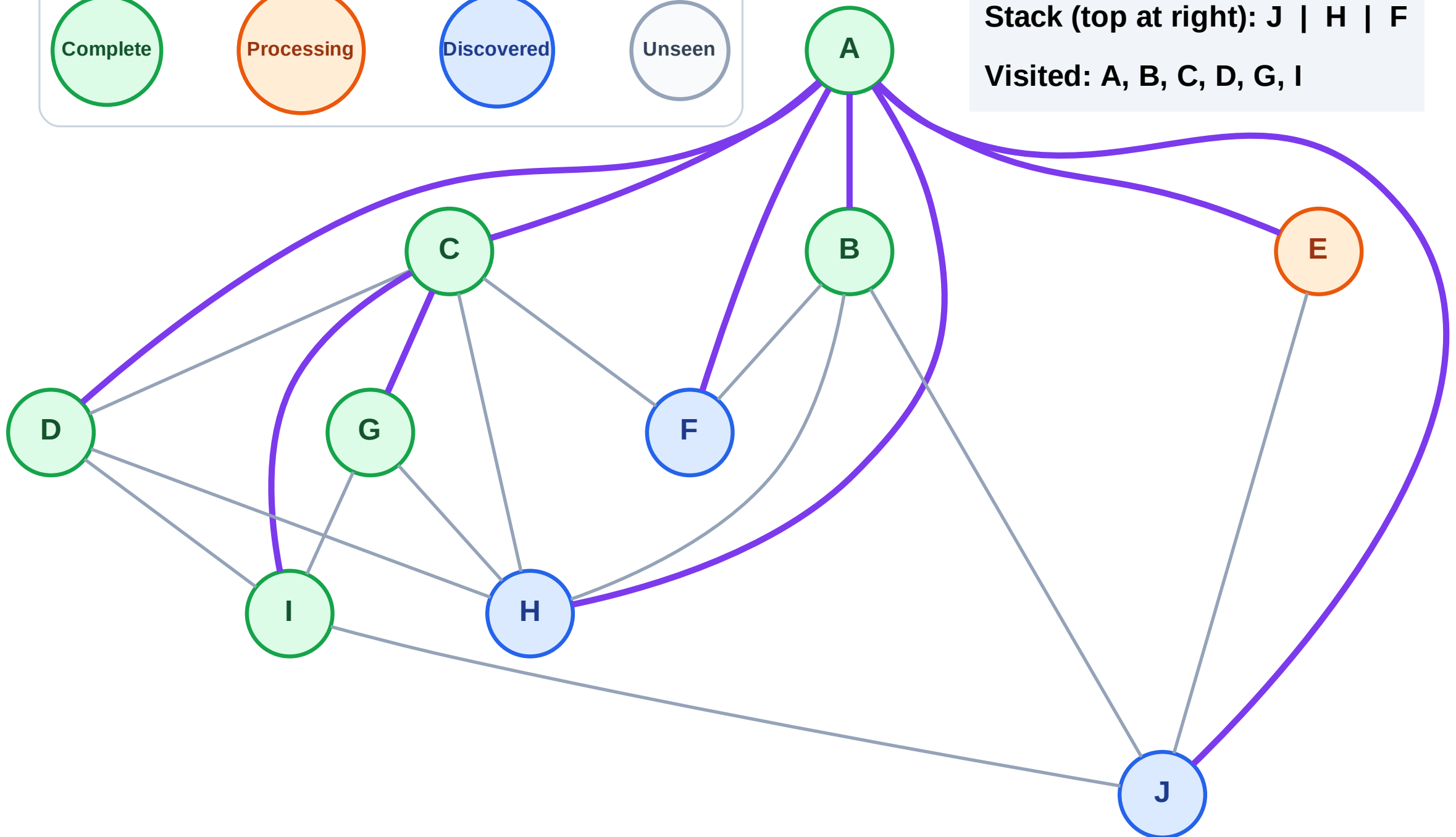
Step 14: Process node E

Legend



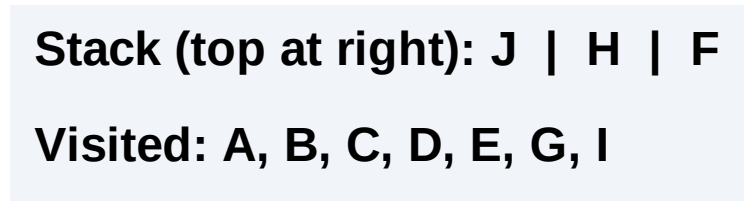
Stack (top at right): J | H | F

Visited: A, B, C, D, G, I



Step 15: No new neighbors from E

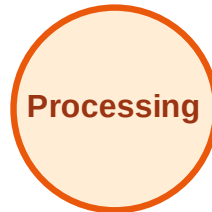
Step 15: No new neighbors from E



DFS Traversal

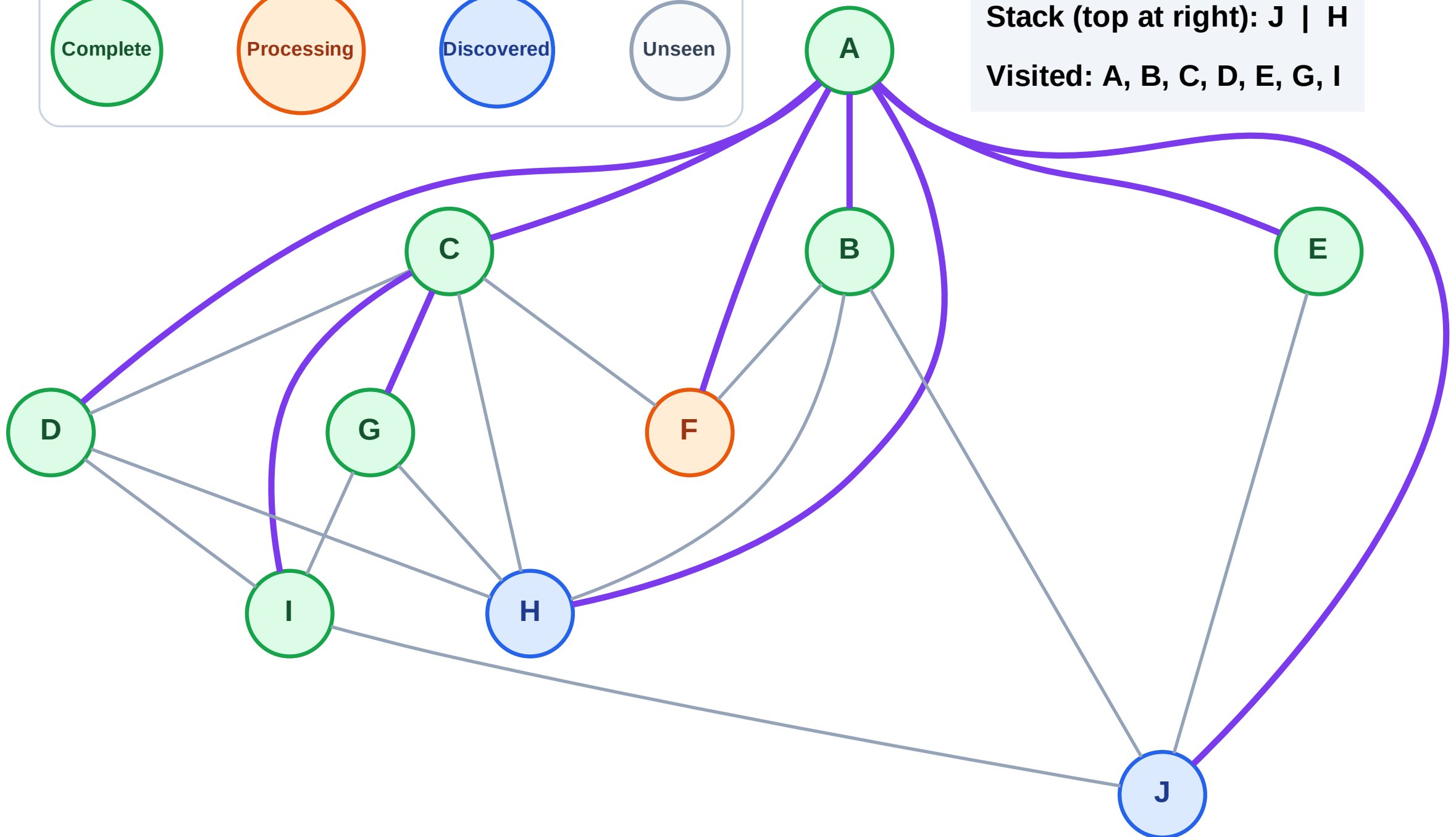
Step 16: Process node F

Legend



Stack (top at right): J | H

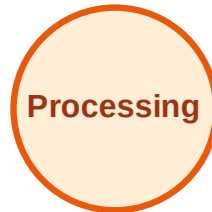
Visited: A, B, C, D, E, G, I



DFS Traversal

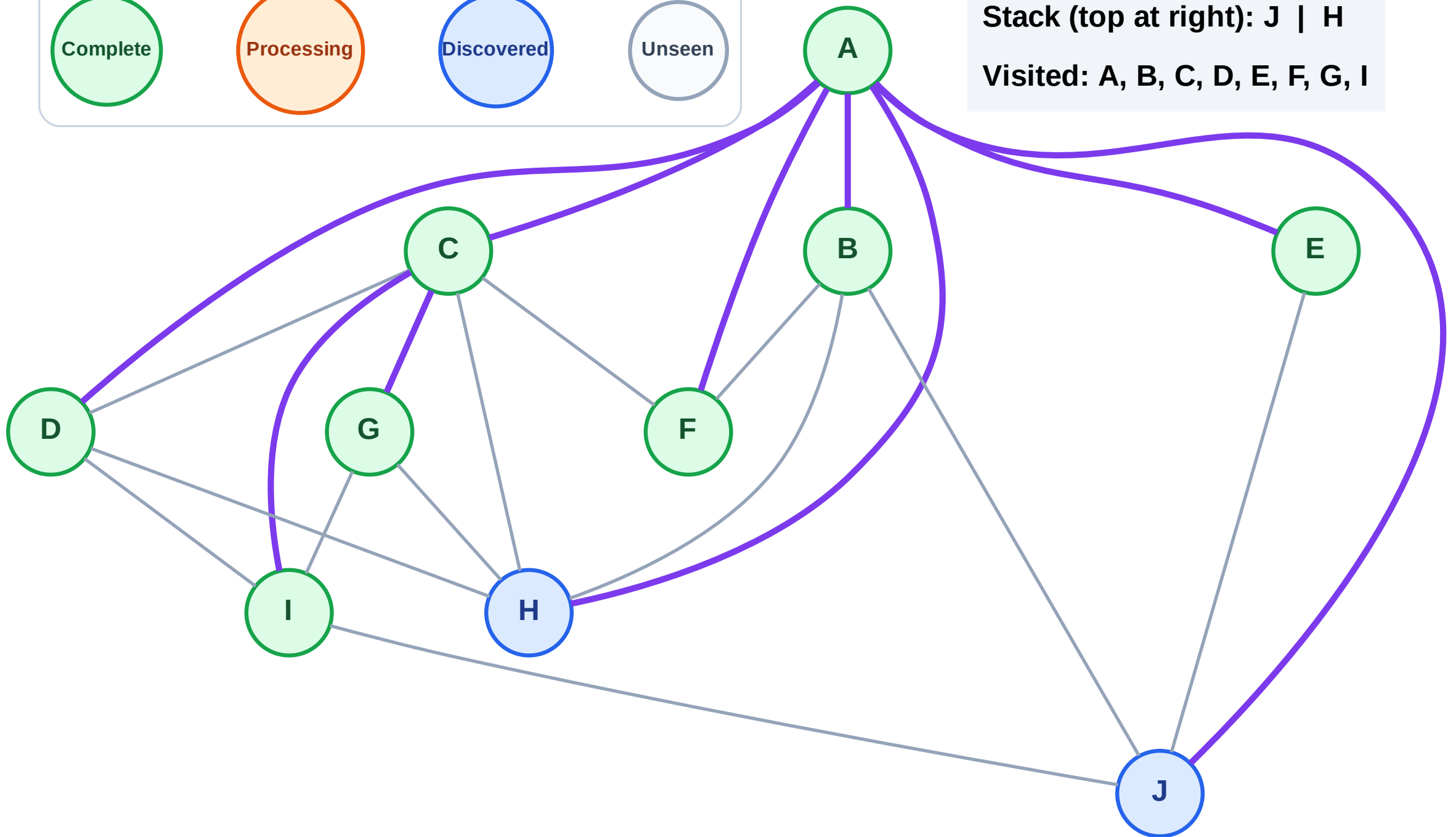
Step 17: No new neighbors from F

Legend



Stack (top at right): J | H

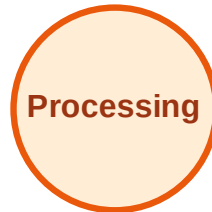
Visited: A, B, C, D, E, F, G, I



DFS Traversal

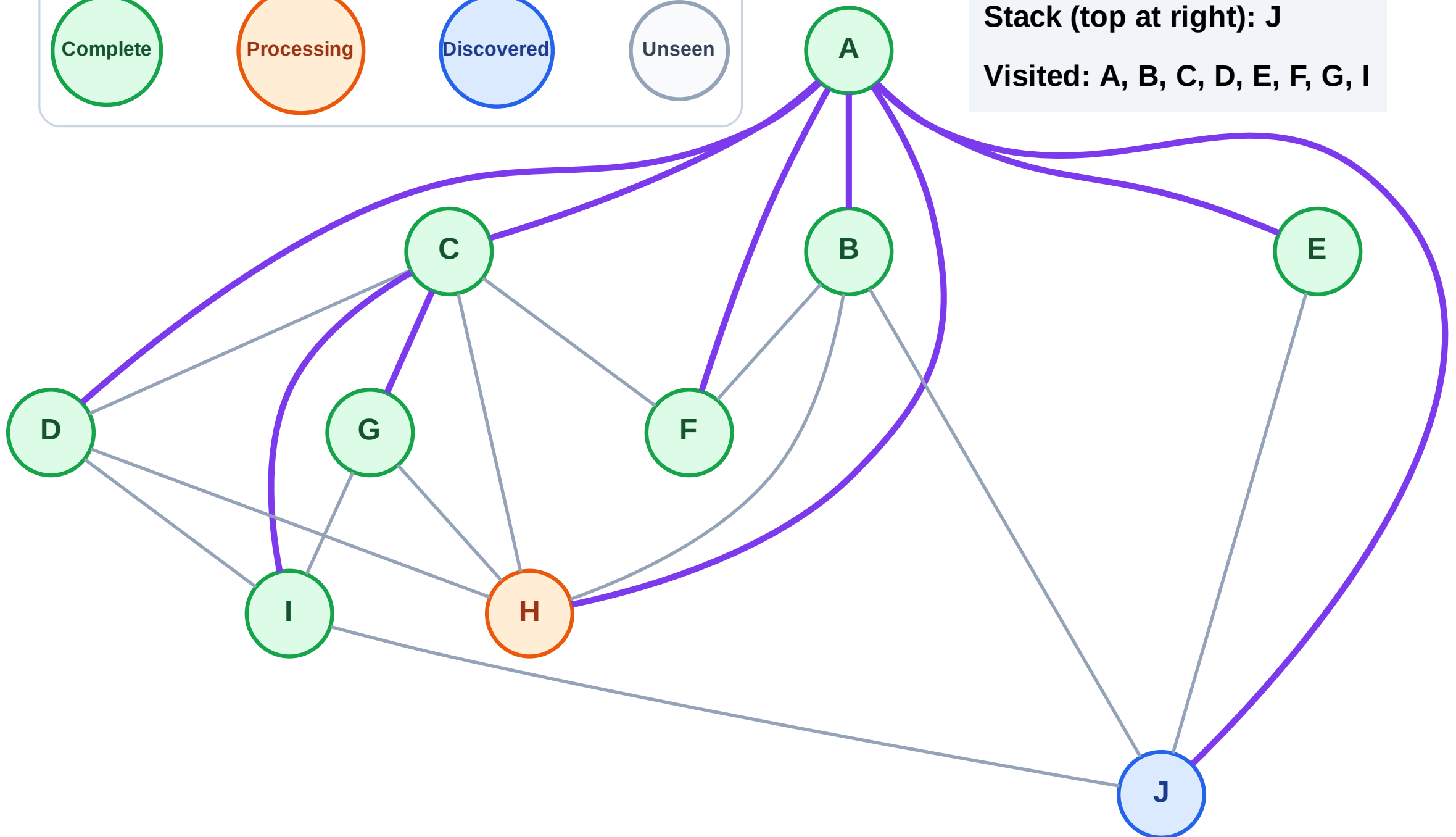
Step 18: Process node H

Legend



Stack (top at right): J

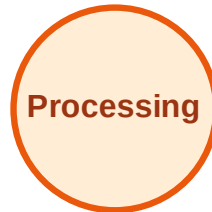
Visited: A, B, C, D, E, F, G, I



DFS Traversal

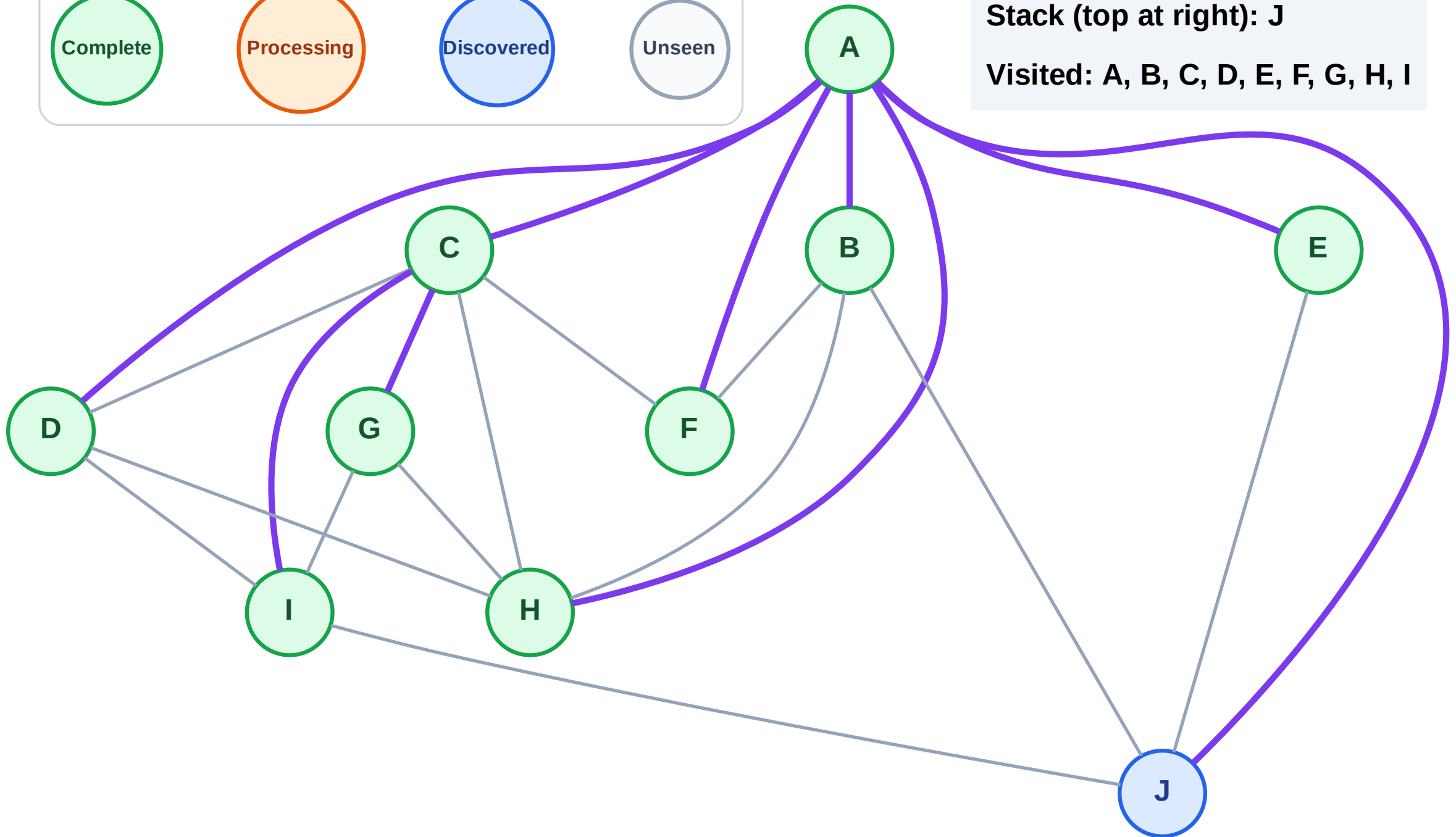
Step 19: No new neighbors from H

Legend



Stack (top at right): J

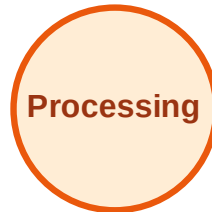
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

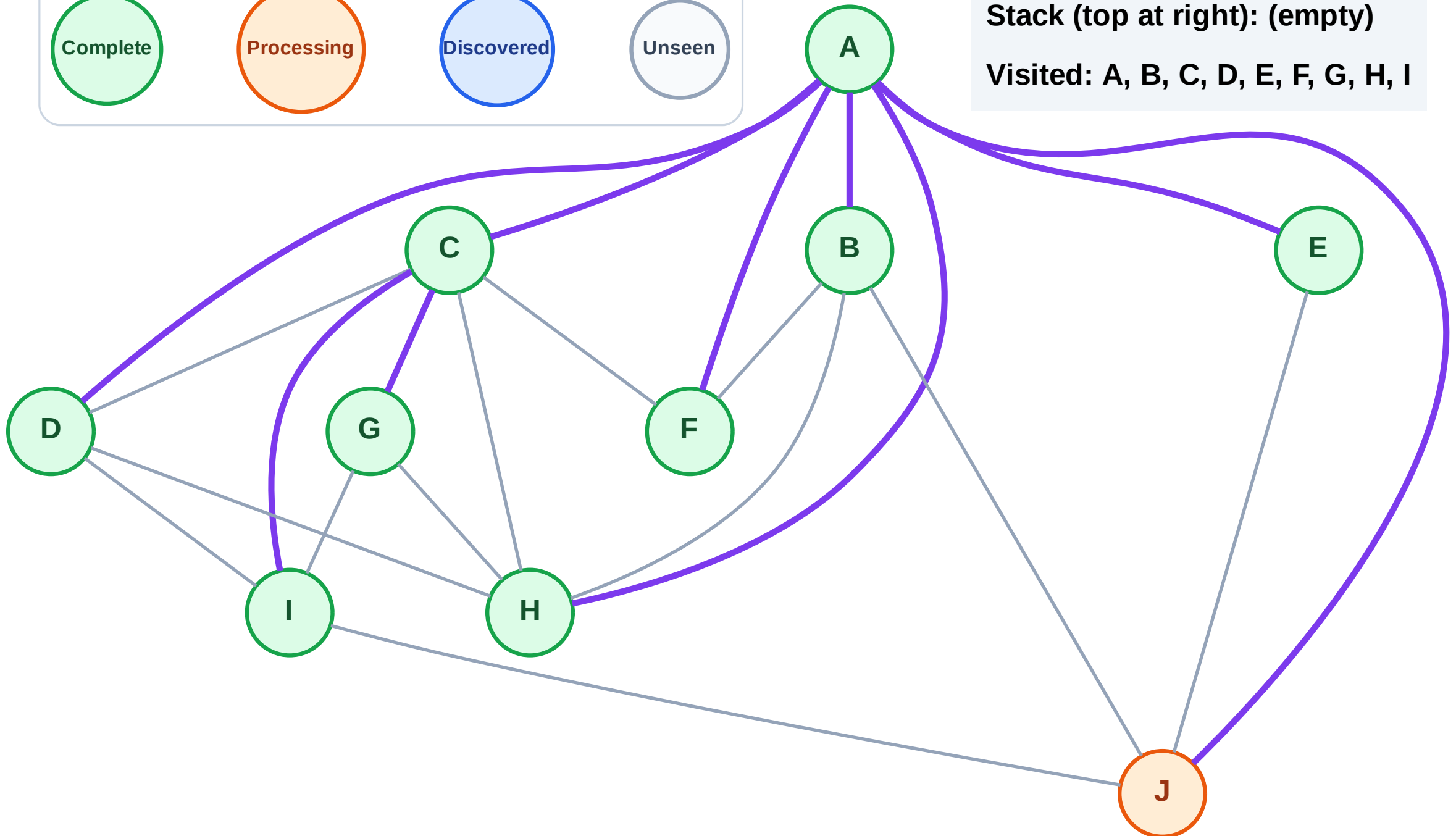
Step 20: Process node J

Legend



Stack (top at right): (empty)

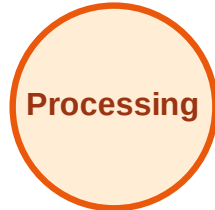
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

